Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:24:03

Measurement Details:

- **Sample:** Muestra8
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

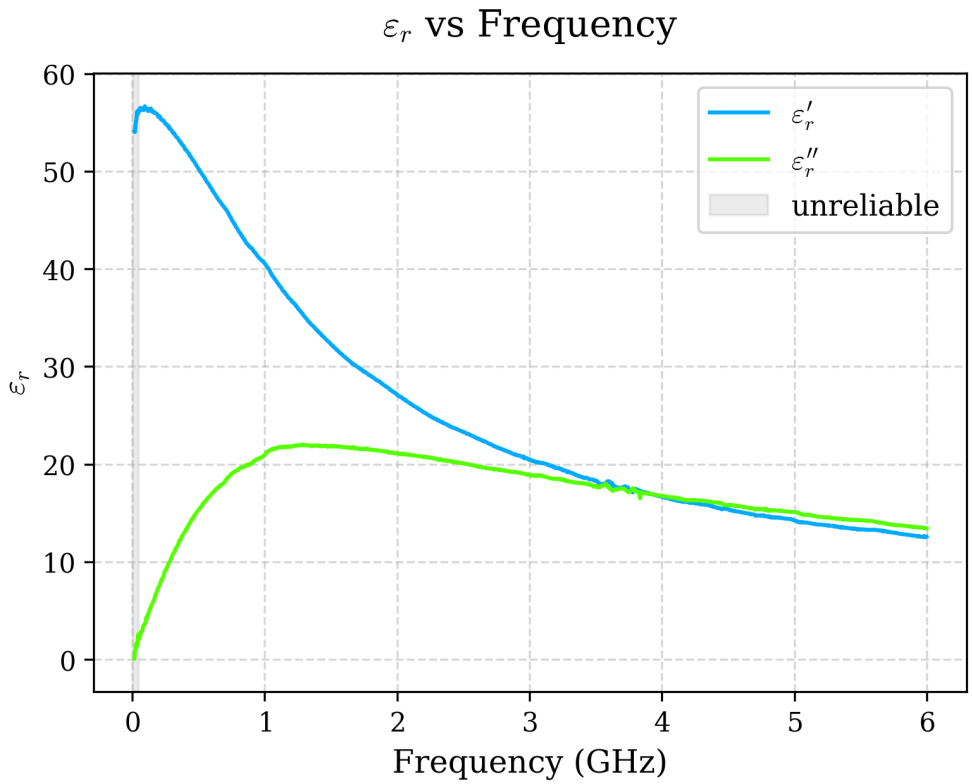
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	54.0760	0.0759	0.0014
0.020	54.0097	0.7740	0.0143
0.025	54.8778	1.1135	0.0203
0.030	55.2437	1.6278	0.0295
0.035	56.1063	1.3329	0.0238
0.040	55.7962	1.8118	0.0325
0.045	56.2438	2.5630	0.0456
0.050	56.0901	2.0356	0.0363
0.055	56.1957	2.2444	0.0399
0.060	56.4930	2.4727	0.0438
0.065	56.3312	2.8424	0.0505
0.070	56.3092	2.8449	0.0505
0.075	56.2693	2.8637	0.0509
0.080	56.3455	3.3371	0.0592
0.085	56.4999	3.5067	0.0621

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	56.2674	3.5494	0.0631
0.095	56.6610	3.6066	0.0637
0.100	56.4494	3.8557	0.0683
0.105	56.4364	4.2941	0.0761
0.110	56.3651	4.1460	0.0736
0.115	56.3233	4.5279	0.0804
0.120	56.1590	4.5643	0.0813
0.125	56.3963	4.8383	0.0858
0.130	56.2266	5.0801	0.0904
0.135	56.3616	5.1570	0.0915
0.140	56.4258	5.4503	0.0966
0.145	56.1695	5.5023	0.0980
0.150	56.0940	5.6819	0.1013
0.155	56.1512	5.8104	0.1035
0.160	56.0092	6.0237	0.1075
0.165	56.0816	6.1645	0.1099
0.170	55.9604	6.3122	0.1128
0.175	55.9569	6.6596	0.1190
0.180	55.9072	6.7649	0.1210
0.185	55.9040	6.9058	0.1235
0.190	55.7650	7.1460	0.1281
0.195	55.5898	7.3228	0.1317
0.200	55.5579	7.4272	0.1337
0.205	55.6505	7.5156	0.1351
0.210	55.5334	7.8000	0.1405
0.215	55.3152	7.8957	0.1427
0.220	55.3271	8.1927	0.1481
0.225	55.2362	8.2086	0.1486
0.230	55.1922	8.3471	0.1512
0.235	55.2037	8.5006	0.1540
0.240	55.0857	8.7241	0.1584
0.245	54.9892	9.0209	0.1640
0.250	54.8555	9.1500	0.1668
0.255	54.8622	9.1439	0.1667
0.260	54.8045	9.3663	0.1709
0.265	54.6494	9.5204	0.1742
0.270	54.5679	9.7034	0.1778
0.275	54.6355	9.8738	0.1807
0.280	54.3496	9.9607	0.1833
0.285	54.3886	10.0511	0.1848
0.290	54.3034	10.2346	0.1885
0.295	54.2125	10.4297	0.1924
0.300	54.1072	10.5539	0.1951
0.305	53.9781	10.7501	0.1992
0.310	53.8804	10.8853	0.2020
0.315	53.9054	10.9204	0.2026

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	53.7952	11.0414	0.2052
0.325	53.6621	11.3101	0.2108
0.330	53.5544	11.3935	0.2127
0.335	53.4744	11.4915	0.2149
0.340	53.4188	11.6239	0.2176
0.345	53.3621	11.7094	0.2194
0.350	53.2335	11.9241	0.2240
0.355	53.1118	12.1612	0.2290
0.360	53.0437	12.2018	0.2300
0.365	52.9913	12.2990	0.2321
0.370	52.8264	12.4301	0.2353
0.375	52.7394	12.6223	0.2393
0.380	52.6835	12.6947	0.2410
0.385	52.6073	12.8459	0.2442
0.390	52.5259	12.9551	0.2466
0.395	52.3363	13.0379	0.2491
0.400	52.2289	13.1304	0.2514
0.405	52.1985	13.3223	0.2552
0.410	52.1104	13.3336	0.2559
0.415	51.9996	13.5197	0.2600
0.420	51.9215	13.6532	0.2630
0.425	51.8477	13.7541	0.2653
0.430	51.7521	13.8908	0.2684
0.435	51.5508	13.9773	0.2711
0.440	51.5317	14.0806	0.2732
0.445	51.3966	14.2707	0.2777
0.450	51.2999	14.3516	0.2798
0.455	51.1615	14.3548	0.2806
0.460	51.1341	14.5498	0.2845
0.465	51.0232	14.6233	0.2866
0.470	50.9072	14.7685	0.2901
0.475	50.7964	14.8326	0.2920
0.480	50.7105	14.9416	0.2946
0.485	50.5705	15.0155	0.2969
0.490	50.4698	15.1385	0.3000
0.495	50.3599	15.2301	0.3024
0.500	50.2378	15.2803	0.3042
0.505	50.1731	15.3893	0.3067
0.510	50.0422	15.4689	0.3091
0.515	49.9155	15.6033	0.3126
0.520	49.8712	15.6182	0.3132
0.525	49.7384	15.7286	0.3162
0.530	49.6273	15.8645	0.3197
0.535	49.5465	15.9466	0.3219
0.540	49.4418	16.0501	0.3246
0.545	49.3454	16.1336	0.3270

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	49.2171	16.1573	0.3283
0.555	49.1186	16.2638	0.3311
0.560	48.9490	16.3363	0.3337
0.565	48.8728	16.4249	0.3361
0.570	48.7920	16.5381	0.3390
0.575	48.7274	16.5843	0.3403
0.580	48.6187	16.6784	0.3430
0.585	48.4742	16.7336	0.3452
0.590	48.3717	16.7760	0.3468
0.595	48.2664	16.9183	0.3505
0.600	48.1935	16.9884	0.3525
0.605	48.0383	17.0428	0.3548
0.610	47.9262	17.1172	0.3572
0.615	47.8503	17.1899	0.3592
0.620	47.7395	17.2166	0.3606
0.625	47.6243	17.3410	0.3641
0.630	47.5438	17.3498	0.3649
0.635	47.4113	17.4825	0.3687
0.640	47.3305	17.5105	0.3700
0.645	47.1799	17.5668	0.3723
0.650	47.1001	17.6344	0.3744
0.655	46.9989	17.7204	0.3770
0.660	46.9245	17.7226	0.3777
0.665	46.8675	17.7921	0.3796
0.670	46.7419	17.8871	0.3827
0.675	46.6617	17.8792	0.3832
0.680	46.5248	17.9644	0.3861
0.685	46.4705	17.9918	0.3872
0.690	46.4071	18.0407	0.3887
0.695	46.2894	18.1262	0.3916
0.700	46.1851	18.2343	0.3948
0.705	46.1532	18.3793	0.3982
0.710	46.0399	18.3933	0.3995
0.715	45.9290	18.5181	0.4032
0.720	45.8387	18.6356	0.4065
0.725	45.6890	18.6778	0.4088
0.730	45.5925	18.7189	0.4106
0.735	45.3952	18.7998	0.4141
0.740	45.3301	18.8854	0.4166
0.745	45.2310	18.9695	0.4194
0.750	45.0255	19.0230	0.4225
0.755	44.9319	19.0471	0.4239
0.760	44.8554	19.1342	0.4266
0.765	44.7314	19.1784	0.4287
0.770	44.5986	19.2169	0.4309
0.775	44.5575	19.2156	0.4313

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	44.3925	19.3132	0.4351
0.785	44.2546	19.3362	0.4369
0.790	44.1796	19.3224	0.4374
0.795	44.0628	19.3885	0.4400
0.800	43.9290	19.3955	0.4415
0.805	43.8119	19.5156	0.4454
0.810	43.7076	19.5387	0.4470
0.815	43.6600	19.6214	0.4494
0.820	43.5213	19.6523	0.4516
0.825	43.3906	19.6129	0.4520
0.830	43.3440	19.6811	0.4541
0.835	43.2040	19.7191	0.4564
0.840	43.0842	19.6962	0.4572
0.845	43.0511	19.7373	0.4585
0.850	42.8917	19.7679	0.4609
0.855	42.7982	19.8593	0.4640
0.860	42.6749	19.8214	0.4645
0.865	42.5707	19.9332	0.4682
0.870	42.5320	19.9564	0.4692
0.875	42.4940	19.9073	0.4685
0.880	42.3431	19.8876	0.4697
0.885	42.2586	19.9440	0.4720
0.890	42.2591	20.0428	0.4743
0.895	42.1415	20.0210	0.4751
0.900	42.1536	20.0395	0.4754
0.905	42.0720	20.1656	0.4793
0.910	41.9672	20.2149	0.4817
0.915	41.8107	20.2657	0.4847
0.920	41.7499	20.2630	0.4853
0.925	41.7127	20.3939	0.4889
0.930	41.6198	20.3933	0.4900
0.935	41.4831	20.4352	0.4926
0.940	41.4284	20.5102	0.4951
0.945	41.3159	20.5167	0.4966
0.950	41.2531	20.5045	0.4970
0.955	41.1715	20.5741	0.4997
0.960	41.0689	20.5829	0.5012
0.965	41.0164	20.6610	0.5037
0.970	40.9655	20.6577	0.5043
0.975	40.8803	20.6607	0.5054
0.980	40.8538	20.7416	0.5077
0.985	40.8189	20.7586	0.5086
0.990	40.6877	20.8247	0.5118
0.995	40.6410	20.8947	0.5141
1.000	40.5638	20.8978	0.5152
1.005	40.4651	21.0477	0.5201

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	40.3926	21.1469	0.5235
1.015	40.2400	21.1587	0.5258
1.020	40.1645	21.2471	0.5290
1.025	40.1160	21.3148	0.5313
1.030	39.9762	21.3393	0.5338
1.035	39.8892	21.3913	0.5363
1.040	39.7499	21.4457	0.5395
1.045	39.6157	21.4613	0.5417
1.050	39.4938	21.5227	0.5450
1.055	39.3267	21.4702	0.5459
1.060	39.2706	21.5768	0.5494
1.065	39.1624	21.4940	0.5488
1.070	39.1005	21.5709	0.5517
1.075	38.9572	21.6177	0.5549
1.080	38.8864	21.5933	0.5553
1.085	38.8006	21.5906	0.5564
1.090	38.7149	21.6564	0.5594
1.095	38.5828	21.6872	0.5621
1.100	38.5144	21.7191	0.5639
1.105	38.4052	21.6877	0.5647
1.110	38.2799	21.7281	0.5676
1.115	38.2154	21.7565	0.5693
1.120	38.1551	21.7326	0.5696
1.125	38.0666	21.7334	0.5709
1.130	37.9344	21.7784	0.5741
1.135	37.7943	21.7514	0.5755
1.140	37.7275	21.6965	0.5751
1.145	37.6461	21.7783	0.5785
1.150	37.6343	21.7364	0.5776
1.155	37.4991	21.7969	0.5813
1.160	37.4014	21.7376	0.5812
1.165	37.3359	21.8006	0.5839
1.170	37.2538	21.7632	0.5842
1.175	37.0957	21.7793	0.5871
1.180	37.0575	21.8226	0.5889
1.185	37.0095	21.7709	0.5883
1.190	36.9042	21.7591	0.5896
1.195	36.8429	21.8115	0.5920
1.200	36.7874	21.8415	0.5937
1.205	36.6469	21.7631	0.5939
1.210	36.6273	21.8291	0.5960
1.215	36.5209	21.8387	0.5980
1.220	36.5093	21.8387	0.5982
1.225	36.4227	21.8623	0.6002
1.230	36.3479	21.9142	0.6029
1.235	36.2047	21.8853	0.6045

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	36.1930	21.9267	0.6058
1.245	36.0988	21.9435	0.6079
1.250	35.9513	21.9175	0.6096
1.255	35.8688	21.9141	0.6110
1.260	35.8256	21.9135	0.6117
1.265	35.6996	21.9550	0.6150
1.270	35.6447	21.9382	0.6155
1.275	35.5726	21.9494	0.6170
1.280	35.4738	21.9511	0.6188
1.285	35.4264	22.0159	0.6215
1.290	35.3013	21.9202	0.6209
1.295	35.1922	21.9461	0.6236
1.300	35.1260	21.9588	0.6251
1.305	35.0571	21.9450	0.6260
1.310	34.9635	21.9527	0.6279
1.315	34.8983	21.9538	0.6291
1.320	34.7945	21.8982	0.6294
1.325	34.7032	21.9220	0.6317
1.330	34.6093	21.9134	0.6332
1.335	34.5430	21.9007	0.6340
1.340	34.5138	21.9094	0.6348
1.345	34.3862	21.8774	0.6362
1.350	34.3125	21.8980	0.6382
1.355	34.2547	21.8216	0.6370
1.360	34.2148	21.8936	0.6399
1.365	34.1338	21.8904	0.6413
1.370	34.0643	21.8782	0.6423
1.375	34.0352	21.9045	0.6436
1.380	33.9118	21.8903	0.6455
1.385	33.8461	21.9087	0.6473
1.390	33.7732	21.8805	0.6479
1.395	33.6981	21.9206	0.6505
1.400	33.6751	21.9107	0.6507
1.405	33.5383	21.8901	0.6527
1.410	33.5235	21.9101	0.6536
1.415	33.4214	21.8328	0.6533
1.420	33.3809	21.8897	0.6558
1.425	33.3242	21.8169	0.6547
1.430	33.1950	21.8527	0.6583
1.435	33.1530	21.8309	0.6585
1.440	33.0759	21.8442	0.6604
1.445	33.0151	21.8455	0.6617
1.450	32.9817	21.8713	0.6631
1.455	32.8802	21.8683	0.6651
1.460	32.8034	21.9097	0.6679
1.465	32.7397	21.8841	0.6684

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	32.6708	21.8446	0.6686
1.475	32.6136	21.7658	0.6674
1.480	32.5778	21.8434	0.6705
1.485	32.4923	21.8272	0.6718
1.490	32.4502	21.8386	0.6730
1.495	32.3549	21.8196	0.6744
1.500	32.2700	21.8463	0.6770
1.505	32.2041	21.8457	0.6784
1.510	32.1366	21.7798	0.6777
1.515	32.1183	21.8913	0.6816
1.520	31.9915	21.8524	0.6831
1.525	31.9465	21.8639	0.6844
1.530	31.9065	21.8123	0.6836
1.535	31.8425	21.8095	0.6849
1.540	31.7550	21.7959	0.6864
1.545	31.6648	21.8391	0.6897
1.550	31.6564	21.8174	0.6892
1.555	31.5925	21.7811	0.6894
1.560	31.5101	21.8073	0.6921
1.565	31.4260	21.8138	0.6941
1.570	31.3737	21.7706	0.6939
1.575	31.3124	21.8064	0.6964
1.580	31.3055	21.7584	0.6950
1.585	31.2122	21.7337	0.6963
1.590	31.1330	21.7503	0.6986
1.595	31.0803	21.7361	0.6994
1.600	31.0172	21.7348	0.7007
1.605	30.9874	21.7359	0.7014
1.610	30.9260	21.6858	0.7012
1.615	30.8529	21.7158	0.7038
1.620	30.7912	21.7508	0.7064
1.625	30.7414	21.7528	0.7076
1.630	30.6751	21.6687	0.7064
1.635	30.6195	21.7177	0.7093
1.640	30.5827	21.7210	0.7102
1.645	30.5037	21.7080	0.7117
1.650	30.4313	21.6288	0.7107
1.655	30.4018	21.6331	0.7116
1.660	30.2902	21.6841	0.7159
1.665	30.3012	21.7591	0.7181
1.670	30.2289	21.7077	0.7181
1.675	30.1528	21.6500	0.7180
1.680	30.1625	21.6825	0.7189
1.685	30.1128	21.5644	0.7161
1.690	30.0541	21.6511	0.7204
1.695	30.0283	21.6140	0.7198

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	29.9622	21.6033	0.7210
1.705	29.8942	21.6076	0.7228
1.710	29.8682	21.5828	0.7226
1.715	29.8571	21.5203	0.7208
1.720	29.7441	21.5657	0.7250
1.725	29.7159	21.5338	0.7247
1.730	29.7189	21.5691	0.7258
1.735	29.6201	21.5602	0.7279
1.740	29.5659	21.5541	0.7290
1.745	29.4656	21.5537	0.7315
1.750	29.5068	21.4990	0.7286
1.755	29.4673	21.5588	0.7316
1.760	29.4149	21.5370	0.7322
1.765	29.3899	21.5677	0.7338
1.770	29.3034	21.5366	0.7350
1.775	29.2661	21.6076	0.7383
1.780	29.2260	21.5224	0.7364
1.785	29.1316	21.5320	0.7391
1.790	29.1249	21.4919	0.7379
1.795	29.0999	21.5097	0.7392
1.800	29.0561	21.4603	0.7386
1.805	28.9687	21.4563	0.7407
1.810	28.9543	21.4525	0.7409
1.815	28.9067	21.4435	0.7418
1.820	28.8708	21.4414	0.7427
1.825	28.8234	21.4455	0.7440
1.830	28.7337	21.4227	0.7456
1.835	28.6959	21.4429	0.7472
1.840	28.6423	21.4321	0.7483
1.845	28.6106	21.4221	0.7487
1.850	28.5628	21.4092	0.7495
1.855	28.5358	21.3953	0.7498
1.860	28.4859	21.3511	0.7495
1.865	28.4703	21.3746	0.7508
1.870	28.4232	21.3793	0.7522
1.875	28.3417	21.3755	0.7542
1.880	28.3148	21.3496	0.7540
1.885	28.2344	21.3181	0.7550
1.890	28.2062	21.3143	0.7557
1.895	28.1503	21.3091	0.7570
1.900	28.0820	21.3446	0.7601
1.905	28.0565	21.3034	0.7593
1.910	28.0054	21.2823	0.7599
1.915	27.9363	21.2937	0.7622
1.920	27.9053	21.2589	0.7618
1.925	27.8666	21.2846	0.7638

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	27.8346	21.2025	0.7617
1.935	27.7803	21.2445	0.7647
1.940	27.7302	21.1450	0.7625
1.945	27.6257	21.2180	0.7681
1.950	27.5916	21.1966	0.7682
1.955	27.5481	21.2078	0.7698
1.960	27.5283	21.1655	0.7689
1.965	27.4420	21.1483	0.7707
1.970	27.4234	21.1390	0.7708
1.975	27.3593	21.1822	0.7742
1.980	27.3015	21.1068	0.7731
1.985	27.3100	21.0938	0.7724
1.990	27.2330	21.1048	0.7750
1.995	27.1775	21.0972	0.7763
2.000	27.0776	21.1059	0.7795
2.005	27.1025	21.0768	0.7777
2.010	27.0043	21.1220	0.7822
2.015	26.9660	21.0284	0.7798
2.020	26.9094	21.0301	0.7815
2.025	26.8841	21.0649	0.7835
2.030	26.8905	21.0333	0.7822
2.035	26.8119	21.0529	0.7852
2.040	26.7801	21.0401	0.7857
2.045	26.6688	20.9796	0.7867
2.050	26.6581	21.0857	0.7910
2.055	26.6435	21.0450	0.7899
2.060	26.5413	21.0288	0.7923
2.065	26.5638	21.0346	0.7919
2.070	26.4748	20.9769	0.7923
2.075	26.4131	20.9439	0.7929
2.080	26.3758	20.9710	0.7951
2.085	26.3374	20.9679	0.7961
2.090	26.3082	20.9691	0.7971
2.095	26.2585	20.9336	0.7972
2.100	26.1691	20.9833	0.8018
2.105	26.1581	20.9445	0.8007
2.110	26.1398	20.9497	0.8014
2.115	26.0584	20.9180	0.8027
2.120	26.0106	20.8677	0.8023
2.125	25.9481	20.8858	0.8049
2.130	25.9270	20.8679	0.8049
2.135	25.8974	20.8747	0.8061
2.140	25.8632	20.8680	0.8069
2.145	25.7951	20.8346	0.8077
2.150	25.8000	20.8360	0.8076
2.155	25.7227	20.8141	0.8092

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	25.6605	20.8101	0.8110
2.165	25.6564	20.8255	0.8117
2.170	25.5742	20.8033	0.8134
2.175	25.5547	20.8234	0.8149
2.180	25.5107	20.8008	0.8154
2.185	25.4776	20.7990	0.8164
2.190	25.4182	20.7743	0.8173
2.195	25.3207	20.7365	0.8190
2.200	25.3003	20.7472	0.8200
2.205	25.2930	20.7031	0.8185
2.210	25.2500	20.7552	0.8220
2.215	25.2005	20.7125	0.8219
2.220	25.1703	20.7127	0.8229
2.225	25.0984	20.7374	0.8262
2.230	25.0697	20.7105	0.8261
2.235	25.0410	20.7060	0.8269
2.240	25.0220	20.6610	0.8257
2.245	24.9271	20.6826	0.8297
2.250	24.9235	20.6451	0.8283
2.255	24.9035	20.6420	0.8289
2.260	24.8141	20.6467	0.8321
2.265	24.7900	20.6491	0.8330
2.270	24.7816	20.5735	0.8302
2.275	24.6984	20.5715	0.8329
2.280	24.6882	20.5456	0.8322
2.285	24.6553	20.5929	0.8352
2.290	24.6122	20.5456	0.8348
2.295	24.6390	20.5276	0.8331
2.300	24.5839	20.4653	0.8325
2.305	24.4835	20.5283	0.8385
2.310	24.5117	20.4373	0.8338
2.315	24.4907	20.4841	0.8364
2.320	24.4244	20.4805	0.8385
2.325	24.3724	20.4028	0.8371
2.330	24.3367	20.5161	0.8430
2.335	24.3029	20.4503	0.8415
2.340	24.3307	20.4052	0.8387
2.345	24.2193	20.4183	0.8431
2.350	24.1926	20.4228	0.8442
2.355	24.1850	20.3462	0.8413
2.360	24.1454	20.4050	0.8451
2.365	24.1125	20.4001	0.8460
2.370	24.0902	20.3485	0.8447
2.375	24.0822	20.3092	0.8433
2.380	24.0298	20.3590	0.8472
2.385	23.9651	20.3386	0.8487

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	23.9792	20.3429	0.8484
2.395	23.8868	20.3503	0.8519
2.400	23.8501	20.3152	0.8518
2.405	23.8507	20.2557	0.8493
2.410	23.8287	20.2819	0.8512
2.415	23.7706	20.2605	0.8523
2.420	23.7379	20.2891	0.8547
2.425	23.7171	20.2204	0.8526
2.430	23.6932	20.2051	0.8528
2.435	23.6975	20.2092	0.8528
2.440	23.6408	20.1944	0.8542
2.445	23.6141	20.1973	0.8553
2.450	23.6063	20.1999	0.8557
2.455	23.5572	20.1563	0.8556
2.460	23.5321	20.1583	0.8566
2.465	23.5107	20.1849	0.8585
2.470	23.4649	20.1167	0.8573
2.475	23.4395	20.1144	0.8581
2.480	23.3980	20.1119	0.8596
2.485	23.3886	20.1302	0.8607
2.490	23.3521	20.0963	0.8606
2.495	23.3018	20.0911	0.8622
2.500	23.3191	20.0779	0.8610
2.505	23.2746	20.0472	0.8613
2.510	23.2455	20.0468	0.8624
2.515	23.2077	20.0293	0.8630
2.520	23.1746	20.0373	0.8646
2.525	23.1133	19.9934	0.8650
2.530	23.1042	19.9871	0.8651
2.535	23.1112	20.0017	0.8655
2.540	23.0670	19.9555	0.8651
2.545	23.0413	19.9330	0.8651
2.550	22.9884	19.9367	0.8672
2.555	22.9884	19.9261	0.8668
2.560	22.9324	19.9262	0.8689
2.565	22.9049	19.9044	0.8690
2.570	22.8876	19.8884	0.8690
2.575	22.8551	19.8846	0.8700
2.580	22.8099	19.8459	0.8701
2.585	22.7627	19.8801	0.8734
2.590	22.7559	19.8115	0.8706
2.595	22.7076	19.8264	0.8731
2.600	22.6663	19.7908	0.8731
2.605	22.6093	19.7496	0.8735
2.610	22.6474	19.7757	0.8732
2.615	22.6166	19.7184	0.8719

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	22.5448	19.7341	0.8753
2.625	22.5080	19.7065	0.8755
2.630	22.4997	19.7363	0.8772
2.635	22.4995	19.6915	0.8752
2.640	22.4117	19.6812	0.8782
2.645	22.4182	19.6730	0.8775
2.650	22.3456	19.6649	0.8800
2.655	22.3565	19.6313	0.8781
2.660	22.2948	19.6337	0.8806
2.665	22.3250	19.6321	0.8794
2.670	22.2856	19.5828	0.8787
2.675	22.2298	19.5856	0.8811
2.680	22.2099	19.5754	0.8814
2.685	22.1472	19.5557	0.8830
2.690	22.1130	19.5430	0.8838
2.695	22.0926	19.5448	0.8847
2.700	22.0618	19.5435	0.8859
2.705	22.0441	19.5297	0.8859
2.710	21.9921	19.5585	0.8893
2.715	21.9946	19.5055	0.8868
2.720	21.9439	19.5141	0.8893
2.725	21.8831	19.4793	0.8902
2.730	21.8360	19.4699	0.8916
2.735	21.8129	19.5047	0.8942
2.740	21.8335	19.4273	0.8898
2.745	21.7708	19.4927	0.8954
2.750	21.7372	19.4696	0.8957
2.755	21.6927	19.4570	0.8969
2.760	21.7098	19.4640	0.8966
2.765	21.6598	19.4152	0.8964
2.770	21.6510	19.4699	0.8993
2.775	21.6009	19.3931	0.8978
2.780	21.5914	19.3902	0.8981
2.785	21.5550	19.4007	0.9001
2.790	21.5597	19.3648	0.8982
2.795	21.4993	19.3909	0.9019
2.800	21.4778	19.3607	0.9014
2.805	21.4706	19.3668	0.9020
2.810	21.3960	19.3223	0.9031
2.815	21.3876	19.3333	0.9039
2.820	21.3865	19.3191	0.9033
2.825	21.3440	19.2832	0.9034
2.830	21.3255	19.3011	0.9051
2.835	21.3082	19.2887	0.9052
2.840	21.2456	19.2839	0.9077
2.845	21.2236	19.2794	0.9084

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	21.2312	19.2065	0.9046
2.855	21.1960	19.2510	0.9082
2.860	21.1568	19.2037	0.9077
2.865	21.0972	19.2098	0.9105
2.870	21.0748	19.1955	0.9108
2.875	21.0619	19.1746	0.9104
2.880	21.0228	19.2029	0.9134
2.885	21.0303	19.1938	0.9127
2.890	20.9430	19.2203	0.9177
2.895	20.9614	19.1516	0.9137
2.900	20.9605	19.1531	0.9138
2.905	20.9110	19.1846	0.9174
2.910	20.8929	19.1813	0.9181
2.915	20.8804	19.1389	0.9166
2.920	20.8746	19.1349	0.9167
2.925	20.8810	19.0874	0.9141
2.930	20.8344	19.0848	0.9160
2.935	20.7902	19.0732	0.9174
2.940	20.8000	19.0804	0.9173
2.945	20.7513	19.0711	0.9190
2.950	20.7084	19.0349	0.9192
2.955	20.7828	19.0327	0.9158
2.960	20.5868	19.0361	0.9247
2.965	20.6921	18.9845	0.9175
2.970	20.6108	18.9363	0.9188
2.975	20.6028	18.9381	0.9192
2.980	20.6056	18.9622	0.9202
2.985	20.5465	18.9218	0.9209
2.990	20.4909	18.9559	0.9251
2.995	20.5017	18.8844	0.9211
3.000	20.4200	18.8982	0.9255
3.005	20.4619	18.8866	0.9230
3.010	20.4306	18.8077	0.9206
3.015	20.3951	18.8142	0.9225
3.020	20.3178	18.8799	0.9292
3.025	20.3125	18.8266	0.9268
3.030	20.3317	18.8023	0.9248
3.035	20.3213	18.8565	0.9279
3.040	20.2530	18.8277	0.9296
3.045	20.2281	18.8050	0.9297
3.050	20.1664	18.7601	0.9303
3.055	20.1873	18.8579	0.9341
3.060	20.1626	18.8482	0.9348
3.065	20.2215	18.8137	0.9304
3.070	20.1272	18.8510	0.9366
3.075	20.1550	18.8461	0.9351

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	20.1372	18.8586	0.9365
3.085	20.1171	18.7799	0.9335
3.090	20.1157	18.8057	0.9349
3.095	20.1133	18.8027	0.9348
3.100	20.1257	18.7827	0.9333
3.105	20.0956	18.7209	0.9316
3.110	20.0883	18.7509	0.9334
3.115	20.0706	18.7166	0.9325
3.120	20.0405	18.7249	0.9344
3.125	20.0154	18.6807	0.9333
3.130	19.9887	18.6602	0.9335
3.135	19.9424	18.6349	0.9344
3.140	19.9587	18.6265	0.9333
3.145	19.9053	18.5799	0.9334
3.150	19.8967	18.5808	0.9339
3.155	19.8501	18.5734	0.9357
3.160	19.8653	18.5496	0.9338
3.165	19.7788	18.5456	0.9376
3.170	19.8249	18.5107	0.9337
3.175	19.7507	18.4896	0.9361
3.180	19.7476	18.4984	0.9367
3.185	19.6779	18.5053	0.9404
3.190	19.6902	18.4915	0.9391
3.195	19.6288	18.4983	0.9424
3.200	19.5838	18.4674	0.9430
3.205	19.5954	18.5114	0.9447
3.210	19.5717	18.4802	0.9442
3.215	19.5801	18.4673	0.9432
3.220	19.6150	18.4921	0.9428
3.225	19.5184	18.5065	0.9482
3.230	19.5588	18.4134	0.9414
3.235	19.5117	18.4549	0.9458
3.240	19.4589	18.4194	0.9466
3.245	19.4839	18.3799	0.9433
3.250	19.3552	18.4518	0.9533
3.255	19.3500	18.3609	0.9489
3.260	19.3902	18.3706	0.9474
3.265	19.3547	18.4002	0.9507
3.270	19.3238	18.2689	0.9454
3.275	19.2967	18.2816	0.9474
3.280	19.3013	18.3006	0.9482
3.285	19.1639	18.2763	0.9537
3.290	19.2450	18.2211	0.9468
3.295	19.1754	18.2087	0.9496
3.300	19.2153	18.2175	0.9481
3.305	19.1554	18.2293	0.9517

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	19.1190	18.2000	0.9519
3.315	19.0808	18.1691	0.9522
3.320	19.0816	18.2012	0.9539
3.325	19.0929	18.1709	0.9517
3.330	19.0239	18.1784	0.9556
3.335	18.9598	18.1028	0.9548
3.340	18.9717	18.1302	0.9556
3.345	18.9277	18.0849	0.9555
3.350	18.8710	18.0842	0.9583
3.355	18.9006	18.0749	0.9563
3.360	18.8179	18.0928	0.9615
3.365	18.8068	18.0755	0.9611
3.370	18.7612	18.0665	0.9630
3.375	18.8016	18.0939	0.9624
3.380	18.7748	18.0760	0.9628
3.385	18.7335	18.0624	0.9642
3.390	18.7065	18.0350	0.9641
3.395	18.7117	18.0424	0.9642
3.400	18.6280	18.0426	0.9686
3.405	18.6252	18.0075	0.9668
3.410	18.6209	18.0382	0.9687
3.415	18.5626	18.0071	0.9701
3.420	18.5828	17.9753	0.9673
3.425	18.5986	17.9742	0.9664
3.430	18.5917	17.9802	0.9671
3.435	18.5496	18.0127	0.9711
3.440	18.5589	17.9377	0.9665
3.445	18.5067	17.9435	0.9696
3.450	18.4354	17.9024	0.9711
3.455	18.4684	17.9376	0.9713
3.460	18.4362	17.9010	0.9710
3.465	18.4201	17.8689	0.9701
3.470	18.4143	17.8517	0.9694
3.475	18.4296	17.8057	0.9662
3.480	18.3999	17.7505	0.9647
3.485	18.4154	17.6760	0.9598
3.490	18.2977	17.7796	0.9717
3.495	18.2801	17.7708	0.9721
3.500	18.2721	17.6971	0.9685
3.505	18.3021	17.7547	0.9701
3.510	18.2589	17.6340	0.9658
3.515	18.1786	17.6901	0.9731
3.520	18.1437	17.6333	0.9719
3.525	17.9744	17.6798	0.9836
3.530	18.0250	17.7223	0.9832
3.535	17.9877	17.6468	0.9811

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	17.9091	17.7136	0.9891
3.545	17.9580	17.7905	0.9907
3.550	17.9015	17.8009	0.9944
3.555	17.9433	17.7816	0.9910
3.560	17.9645	17.8442	0.9933
3.565	18.0210	17.8316	0.9895
3.570	17.9164	17.8517	0.9964
3.575	17.9229	18.0060	1.0046
3.580	18.0593	17.9565	0.9943
3.585	18.2258	17.8558	0.9797
3.590	18.2400	17.6832	0.9695
3.595	18.2584	17.6048	0.9642
3.600	18.1795	17.6308	0.9698
3.605	18.2116	17.4924	0.9605
3.610	18.1541	17.4546	0.9615
3.615	18.0339	17.4164	0.9658
3.620	17.9605	17.3606	0.9666
3.625	18.0461	17.2598	0.9564
3.630	17.7653	17.2679	0.9720
3.635	17.7641	17.3518	0.9768
3.640	17.7811	17.4053	0.9789
3.645	17.7254	17.4129	0.9824
3.650	17.6552	17.4148	0.9864
3.655	17.6406	17.3892	0.9857
3.660	17.6113	17.4650	0.9917
3.665	17.6175	17.4214	0.9889
3.670	17.5645	17.4531	0.9937
3.675	17.5835	17.4249	0.9910
3.680	17.5832	17.4599	0.9930
3.685	17.5726	17.5043	0.9961
3.690	17.5774	17.4928	0.9952
3.695	17.5808	17.5241	0.9968
3.700	17.6083	17.5187	0.9949
3.705	17.6491	17.5043	0.9918
3.710	17.6179	17.4762	0.9920
3.715	17.6795	17.4720	0.9883
3.720	17.7368	17.4064	0.9814
3.725	17.6925	17.3991	0.9834
3.730	17.7008	17.3211	0.9786
3.735	17.6558	17.2123	0.9749
3.740	17.6423	17.1792	0.9737
3.745	17.5817	17.1468	0.9753
3.750	17.4701	17.1154	0.9797
3.755	17.3881	17.1416	0.9858
3.760	17.2978	17.1852	0.9935
3.765	17.2239	17.2555	1.0018

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	17.2751	17.3811	1.0061
3.775	17.3629	17.3629	1.0000
3.780	17.0947	17.5484	1.0265
3.785	17.3814	17.4238	1.0024
3.790	17.4799	17.3026	0.9899
3.795	17.4525	17.2784	0.9900
3.800	17.4075	17.3391	0.9961
3.805	17.4672	17.2313	0.9865
3.810	17.3853	17.1967	0.9892
3.815	17.3380	17.0943	0.9859
3.820	17.3618	17.1611	0.9884
3.825	17.3117	17.0734	0.9862
3.830	17.3027	17.1126	0.9890
3.835	16.9339	16.4963	0.9742
3.840	17.2320	17.0799	0.9912
3.845	17.1990	17.0817	0.9932
3.850	17.1982	17.0596	0.9919
3.855	17.1535	17.0676	0.9950
3.860	17.1333	17.0174	0.9932
3.865	17.0613	17.0207	0.9976
3.870	17.1294	17.0315	0.9943
3.875	17.0678	17.0136	0.9968
3.880	17.0946	17.0038	0.9947
3.885	17.0326	17.0022	0.9982
3.890	17.0408	17.0271	0.9992
3.895	17.0533	16.9531	0.9941
3.900	17.0088	16.9456	0.9963
3.905	16.9422	16.9764	1.0020
3.910	16.9755	16.9419	0.9980
3.915	16.9514	16.9308	0.9988
3.920	16.9219	16.9537	1.0019
3.925	16.9010	16.9010	1.0000
3.930	16.8950	16.8979	1.0002
3.935	16.9051	16.8842	0.9988
3.940	16.8580	16.8820	1.0014
3.945	16.8494	16.8540	1.0003
3.950	16.8129	16.8393	1.0016
3.955	16.7837	16.8474	1.0038
3.960	16.7845	16.8206	1.0022
3.965	16.7852	16.8181	1.0020
3.970	16.7424	16.8265	1.0050
3.975	16.7259	16.7996	1.0044
3.980	16.7146	16.7835	1.0041
3.985	16.7088	16.7936	1.0051
3.990	16.7034	16.7820	1.0047
3.995	16.6704	16.7507	1.0048

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	16.6440	16.7493	1.0063
4.005	16.6263	16.7382	1.0067
4.010	16.5935	16.7194	1.0076
4.015	16.5913	16.7422	1.0091
4.020	16.5788	16.7061	1.0077
4.025	16.6153	16.6818	1.0040
4.030	16.5318	16.6742	1.0086
4.035	16.5103	16.6876	1.0107
4.040	16.5279	16.6714	1.0087
4.045	16.5214	16.6417	1.0073
4.050	16.4916	16.6149	1.0075
4.055	16.4615	16.6061	1.0088
4.060	16.4296	16.6043	1.0106
4.065	16.3973	16.5893	1.0117
4.070	16.4055	16.6145	1.0127
4.075	16.3848	16.5829	1.0121
4.080	16.3622	16.6115	1.0152
4.085	16.3521	16.5814	1.0140
4.090	16.3527	16.5621	1.0128
4.095	16.2943	16.5720	1.0170
4.100	16.3175	16.5854	1.0164
4.105	16.2887	16.4677	1.0110
4.110	16.2966	16.5370	1.0148
4.115	16.2550	16.5132	1.0159
4.120	16.2215	16.4965	1.0170
4.125	16.2729	16.5032	1.0142
4.130	16.2533	16.5220	1.0165
4.135	16.1941	16.4510	1.0159
4.140	16.2135	16.3986	1.0114
4.145	16.2812	16.4113	1.0080
4.150	16.1440	16.4154	1.0168
4.155	16.1229	16.4133	1.0180
4.160	16.1678	16.4117	1.0151
4.165	16.1208	16.3507	1.0143
4.170	16.1174	16.3409	1.0139
4.175	16.1561	16.3001	1.0089
4.180	16.1536	16.3410	1.0116
4.185	16.1327	16.2825	1.0093
4.190	16.0895	16.2815	1.0119
4.195	16.0882	16.2812	1.0120
4.200	16.1161	16.2966	1.0112
4.205	16.1064	16.2987	1.0119
4.210	16.0260	16.2911	1.0165
4.215	16.0774	16.3036	1.0141
4.220	16.0486	16.3098	1.0163
4.225	16.1105	16.3074	1.0122

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	16.0473	16.2915	1.0152
4.235	16.0097	16.2811	1.0169
4.240	15.9993	16.3436	1.0215
4.245	16.0330	16.3025	1.0168
4.250	15.9984	16.2765	1.0174
4.255	16.0077	16.3083	1.0188
4.260	15.9541	16.2855	1.0208
4.265	16.0020	16.2787	1.0173
4.270	15.9490	16.2696	1.0201
4.275	15.9280	16.2655	1.0212
4.280	15.9256	16.2772	1.0221
4.285	15.9080	16.2763	1.0232
4.290	15.8840	16.2620	1.0238
4.295	15.8824	16.2518	1.0233
4.300	15.8776	16.2714	1.0248
4.305	15.8609	16.2589	1.0251
4.310	15.8567	16.2426	1.0243
4.315	15.8432	16.2417	1.0252
4.320	15.8477	16.2292	1.0241
4.325	15.8266	16.2309	1.0255
4.330	15.8367	16.2231	1.0244
4.335	15.8244	16.2211	1.0251
4.340	15.8285	16.1930	1.0230
4.345	15.7982	16.1781	1.0240
4.350	15.7926	16.1752	1.0242
4.355	15.7753	16.1809	1.0257
4.360	15.7595	16.1490	1.0247
4.365	15.7215	16.1422	1.0268
4.370	15.7484	16.1347	1.0245
4.375	15.6884	16.1365	1.0286
4.380	15.6948	16.1243	1.0274
4.385	15.6227	16.0931	1.0301
4.390	15.6586	16.0686	1.0262
4.395	15.6130	16.0773	1.0297
4.400	15.5944	16.0886	1.0317
4.405	15.5537	16.0729	1.0334
4.410	15.5723	16.0792	1.0326
4.415	15.5529	16.0462	1.0317
4.420	15.5149	16.0875	1.0369
4.425	15.5335	16.0852	1.0355
4.430	15.5068	16.0363	1.0342
4.435	15.4622	15.9971	1.0346
4.440	15.4779	16.0241	1.0353
4.445	15.4093	15.9778	1.0369
4.450	15.4382	15.9389	1.0324
4.455	15.3623	15.9671	1.0394

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	15.4051	15.8844	1.0311
4.465	15.4149	15.8264	1.0267
4.470	15.4266	15.8600	1.0281
4.475	15.4475	15.8167	1.0239
4.480	15.4656	15.8341	1.0238
4.485	15.3834	15.7686	1.0250
4.490	15.4044	15.7993	1.0256
4.495	15.4130	15.7908	1.0245
4.500	15.3778	15.7917	1.0269
4.505	15.3434	15.7711	1.0279
4.510	15.2767	15.7659	1.0320
4.515	15.2845	15.8218	1.0352
4.520	15.2708	15.8064	1.0351
4.525	15.2970	15.8018	1.0330
4.530	15.2291	15.7677	1.0354
4.535	15.2208	15.7894	1.0374
4.540	15.2348	15.7561	1.0342
4.545	15.1922	15.7410	1.0361
4.550	15.1999	15.7042	1.0332
4.555	15.1648	15.7242	1.0369
4.560	15.1687	15.7206	1.0364
4.565	15.1446	15.7335	1.0389
4.570	15.1435	15.6962	1.0365
4.575	15.1379	15.6896	1.0364
4.580	15.1296	15.7068	1.0381
4.585	15.1076	15.6879	1.0384
4.590	15.0989	15.6647	1.0375
4.595	15.0718	15.6735	1.0399
4.600	15.0650	15.6551	1.0392
4.605	15.0378	15.6415	1.0401
4.610	15.0360	15.6376	1.0400
4.615	15.0172	15.6197	1.0401
4.620	15.0299	15.5964	1.0377
4.625	15.0130	15.6092	1.0397
4.630	15.0007	15.5784	1.0385
4.635	14.9761	15.5558	1.0387
4.640	14.9498	15.5545	1.0404
4.645	14.9543	15.5462	1.0396
4.650	14.9353	15.5427	1.0407
4.655	14.9372	15.4892	1.0370
4.660	14.9470	15.5352	1.0394
4.665	14.9321	15.5039	1.0383
4.670	14.8451	15.4920	1.0436
4.675	14.8539	15.4346	1.0391
4.680	14.8512	15.4513	1.0404
4.685	14.8579	15.4358	1.0389

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	14.8464	15.4795	1.0426
4.695	14.8469	15.4078	1.0378
4.700	14.8126	15.4458	1.0427
4.705	14.7863	15.3201	1.0361
4.710	14.6817	15.4160	1.0500
4.715	14.7377	15.4087	1.0455
4.720	14.7296	15.3618	1.0429
4.725	14.7237	15.4211	1.0474
4.730	14.7381	15.3890	1.0442
4.735	14.7491	15.4242	1.0458
4.740	14.7520	15.3959	1.0437
4.745	14.6877	15.3794	1.0471
4.750	14.7470	15.3625	1.0417
4.755	14.7321	15.4021	1.0455
4.760	14.7089	15.3545	1.0439
4.765	14.7629	15.3809	1.0419
4.770	14.7418	15.3916	1.0441
4.775	14.6490	15.2889	1.0437
4.780	14.6809	15.3061	1.0426
4.785	14.6578	15.4504	1.0541
4.790	14.6696	15.2614	1.0403
4.795	14.6185	15.2724	1.0447
4.800	14.6272	15.2797	1.0446
4.805	14.6483	15.2164	1.0388
4.810	14.5963	15.2930	1.0477
4.815	14.6389	15.2030	1.0385
4.820	14.5399	15.2635	1.0498
4.825	14.5907	15.2656	1.0463
4.830	14.5273	15.2614	1.0505
4.835	14.5301	15.2751	1.0513
4.840	14.5420	15.2640	1.0496
4.845	14.5585	15.2403	1.0468
4.850	14.4972	15.2511	1.0520
4.855	14.5456	15.2748	1.0501
4.860	14.5241	15.2400	1.0493
4.865	14.5682	15.2317	1.0455
4.870	14.5119	15.2238	1.0491
4.875	14.5161	15.2415	1.0500
4.880	14.5089	15.2582	1.0516
4.885	14.4989	15.2116	1.0492
4.890	14.5493	15.2598	1.0488
4.895	14.4820	15.2019	1.0497
4.900	14.4903	15.2263	1.0508
4.905	14.4851	15.2095	1.0500
4.910	14.4553	15.1928	1.0510
4.915	14.4505	15.1913	1.0513

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	14.4499	15.1760	1.0502
4.925	14.4429	15.1655	1.0500
4.930	14.4162	15.1504	1.0509
4.935	14.4149	15.1445	1.0506
4.940	14.4156	15.1379	1.0501
4.945	14.3926	15.1362	1.0517
4.950	14.4002	15.1290	1.0506
4.955	14.3973	15.1347	1.0512
4.960	14.3735	15.1096	1.0512
4.965	14.3765	15.1108	1.0511
4.970	14.3650	15.1183	1.0524
4.975	14.3792	15.1085	1.0507
4.980	14.3296	15.1073	1.0543
4.985	14.3240	15.0866	1.0532
4.990	14.3110	15.1012	1.0552
4.995	14.2523	15.0973	1.0593
5.000	14.2537	15.0778	1.0578
5.005	14.2260	15.0829	1.0602
5.010	14.1991	15.0603	1.0606
5.015	14.1797	15.0833	1.0637
5.020	14.1616	15.0321	1.0615
5.025	14.0971	15.0215	1.0656
5.030	14.0583	14.9899	1.0663
5.035	14.0709	15.0022	1.0662
5.040	14.0369	14.9599	1.0658
5.045	14.0469	14.9208	1.0622
5.050	14.0265	14.8573	1.0592
5.055	14.0598	14.8772	1.0581
5.060	14.0344	14.8489	1.0580
5.065	14.0508	14.8134	1.0543
5.070	14.0284	14.8493	1.0585
5.075	14.0288	14.8195	1.0564
5.080	14.0046	14.7933	1.0563
5.085	13.9990	14.7756	1.0555
5.090	14.0048	14.7999	1.0568
5.095	14.0146	14.8152	1.0571
5.100	14.0145	14.8030	1.0563
5.105	13.9807	14.8546	1.0625
5.110	13.9734	14.7732	1.0572
5.115	13.9499	14.7638	1.0583
5.120	13.9479	14.7596	1.0582
5.125	13.9371	14.7641	1.0593
5.130	13.9338	14.7352	1.0575
5.135	13.8962	14.7210	1.0594
5.140	13.8975	14.7373	1.0604
5.145	13.8796	14.6801	1.0577

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	13.8693	14.7215	1.0614
5.155	13.8614	14.6828	1.0593
5.160	13.8530	14.6708	1.0590
5.165	13.8296	14.6620	1.0602
5.170	13.8307	14.6364	1.0583
5.175	13.8087	14.6185	1.0586
5.180	13.8135	14.6156	1.0581
5.185	13.7970	14.6032	1.0584
5.190	13.8040	14.5948	1.0573
5.195	13.7994	14.5825	1.0567
5.200	13.8059	14.5865	1.0565
5.205	13.7995	14.5852	1.0569
5.210	13.7961	14.5737	1.0564
5.215	13.7913	14.5800	1.0572
5.220	13.7734	14.5557	1.0568
5.225	13.7554	14.5519	1.0579
5.230	13.7712	14.5455	1.0562
5.235	13.7426	14.5566	1.0592
5.240	13.7324	14.5352	1.0585
5.245	13.7354	14.5406	1.0586
5.250	13.7101	14.5451	1.0609
5.255	13.6789	14.5370	1.0627
5.260	13.7367	14.5221	1.0572
5.265	13.7253	14.5605	1.0609
5.270	13.6826	14.5165	1.0610
5.275	13.6671	14.4823	1.0596
5.280	13.6797	14.5075	1.0605
5.285	13.6926	14.4887	1.0581
5.290	13.6576	14.4665	1.0592
5.295	13.6927	14.4301	1.0539
5.300	13.6480	14.4557	1.0592
5.305	13.6799	14.4568	1.0568
5.310	13.5807	14.4316	1.0626
5.315	13.6230	14.4370	1.0597
5.320	13.5989	14.3926	1.0584
5.325	13.6123	14.4205	1.0594
5.330	13.5938	14.4079	1.0599
5.335	13.5340	14.4006	1.0640
5.340	13.5300	14.4277	1.0664
5.345	13.5134	14.4433	1.0688
5.350	13.5727	14.3893	1.0602
5.355	13.5057	14.3912	1.0656
5.360	13.5049	14.3207	1.0604
5.365	13.5686	14.3823	1.0600
5.370	13.4769	14.3197	1.0625
5.375	13.5010	14.3610	1.0637

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	13.5129	14.3635	1.0629
5.385	13.4705	14.3217	1.0632
5.390	13.4610	14.3616	1.0669
5.395	13.4978	14.3548	1.0635
5.400	13.4541	14.3359	1.0655
5.405	13.4533	14.3227	1.0646
5.410	13.4383	14.3258	1.0660
5.415	13.4446	14.2774	1.0619
5.420	13.4056	14.3232	1.0685
5.425	13.4100	14.3404	1.0694
5.430	13.4322	14.2976	1.0644
5.435	13.4187	14.2856	1.0646
5.440	13.3669	14.3073	1.0704
5.445	13.4187	14.2971	1.0655
5.450	13.3597	14.2783	1.0688
5.455	13.3917	14.2953	1.0675
5.460	13.3700	14.2718	1.0674
5.465	13.3818	14.3173	1.0699
5.470	13.3159	14.2724	1.0718
5.475	13.3455	14.2571	1.0683
5.480	13.3188	14.2759	1.0719
5.485	13.3316	14.2501	1.0689
5.490	13.2808	14.2793	1.0752
5.495	13.3298	14.2460	1.0687
5.500	13.3040	14.2648	1.0722
5.505	13.3112	14.2729	1.0723
5.510	13.3129	14.2494	1.0703
5.515	13.3196	14.2404	1.0691
5.520	13.2918	14.2288	1.0705
5.525	13.3064	14.2290	1.0693
5.530	13.2949	14.2220	1.0697
5.535	13.3026	14.2112	1.0683
5.540	13.2821	14.2197	1.0706
5.545	13.2593	14.1953	1.0706
5.550	13.2685	14.1919	1.0696
5.555	13.2748	14.1821	1.0683
5.560	13.2654	14.1815	1.0691
5.565	13.2738	14.1804	1.0683
5.570	13.2689	14.1822	1.0688
5.575	13.2757	14.1760	1.0678
5.580	13.2709	14.1891	1.0692
5.585	13.2632	14.1759	1.0688
5.590	13.2766	14.1633	1.0668
5.595	13.2678	14.1500	1.0665
5.600	13.2769	14.1519	1.0659
5.605	13.2928	14.1288	1.0629

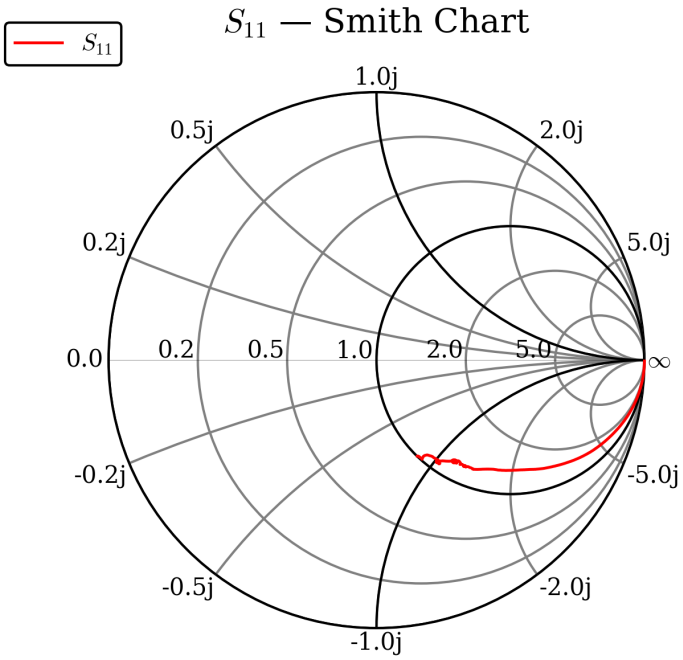
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	13.2625	14.1016	1.0633
5.615	13.2583	14.0691	1.0612
5.620	13.2482	14.0712	1.0621
5.625	13.2415	14.0899	1.0641
5.630	13.2243	14.0462	1.0622
5.635	13.2303	14.0286	1.0603
5.640	13.2220	13.9927	1.0583
5.645	13.1900	13.9826	1.0601
5.650	13.2006	14.0151	1.0617
5.655	13.1758	13.9898	1.0618
5.660	13.1593	13.9938	1.0634
5.665	13.2141	13.9825	1.0582
5.670	13.1068	13.9790	1.0665
5.675	13.1434	13.9342	1.0602
5.680	13.1148	13.9311	1.0622
5.685	13.1563	13.9149	1.0577
5.690	13.1213	13.9072	1.0599
5.695	13.1093	13.9492	1.0641
5.700	13.0938	13.8691	1.0592
5.705	13.0392	13.8636	1.0632
5.710	13.0948	13.8548	1.0580
5.715	13.0615	13.8165	1.0578
5.720	13.0505	13.8355	1.0601
5.725	13.0160	13.8310	1.0626
5.730	13.0138	13.8131	1.0614
5.735	13.0118	13.8038	1.0609
5.740	12.9991	13.8122	1.0626
5.745	12.9997	13.8195	1.0631
5.750	12.9667	13.7853	1.0631
5.755	12.9755	13.7872	1.0626
5.760	12.9342	13.7531	1.0633
5.765	12.9756	13.7510	1.0598
5.770	12.9548	13.7236	1.0594
5.775	12.9443	13.7408	1.0615
5.780	12.9094	13.7400	1.0643
5.785	12.8915	13.7244	1.0646
5.790	12.8726	13.7363	1.0671
5.795	12.8787	13.7373	1.0667
5.800	12.8545	13.7194	1.0673
5.805	12.8486	13.6984	1.0661
5.810	12.8461	13.7162	1.0677
5.815	12.8384	13.7155	1.0683
5.820	12.8326	13.6910	1.0669
5.825	12.8183	13.6835	1.0675
5.830	12.7934	13.6737	1.0688
5.835	12.7985	13.6738	1.0684

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	12.7903	13.6689	1.0687
5.845	12.7724	13.6720	1.0704
5.850	12.7738	13.6536	1.0689
5.855	12.7660	13.6355	1.0681
5.860	12.7448	13.6337	1.0697
5.865	12.7540	13.6375	1.0693
5.870	12.7401	13.6115	1.0684
5.875	12.7275	13.6140	1.0697
5.880	12.7218	13.6020	1.0692
5.885	12.7300	13.5839	1.0671
5.890	12.6739	13.6061	1.0736
5.895	12.6775	13.5911	1.0721
5.900	12.6854	13.5573	1.0687
5.905	12.6727	13.5939	1.0727
5.910	12.6626	13.5274	1.0683
5.915	12.6514	13.5459	1.0707
5.920	12.6463	13.5308	1.0699
5.925	12.6432	13.5140	1.0689
5.930	12.5878	13.5268	1.0746
5.935	12.5840	13.5035	1.0731
5.940	12.5851	13.5095	1.0735
5.945	12.6397	13.4943	1.0676
5.950	12.5711	13.5093	1.0746
5.955	12.5787	13.5149	1.0744
5.960	12.5587	13.5015	1.0751
5.965	12.5551	13.4972	1.0750
5.970	12.5423	13.5153	1.0776
5.975	12.5790	13.4356	1.0681
5.980	12.6379	13.4852	1.0670
5.985	12.5083	13.4730	1.0771
5.990	12.5123	13.4362	1.0738
5.995	12.5180	13.4223	1.0722
6.000	12.5548	13.4168	1.0687

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0035	0.00
0.010	0.9999	-0.0071	-0.00
0.015	0.9999	-0.0108	-0.00
0.020	0.9997	-0.0144	-0.00
0.025	0.9996	-0.0180	-0.00
0.030	0.9993	-0.0218	-0.00
0.035	0.9991	-0.0255	-0.00
0.040	0.9988	-0.0290	-0.01
0.045	0.9984	-0.0326	-0.01
0.050	0.9981	-0.0363	-0.01
0.055	0.9979	-0.0399	-0.01
0.060	0.9974	-0.0435	-0.01
0.065	0.9970	-0.0472	-0.02
0.070	0.9965	-0.0507	-0.02
0.075	0.9961	-0.0542	-0.02
0.080	0.9954	-0.0578	-0.03
0.085	0.9948	-0.0616	-0.03
0.090	0.9942	-0.0649	-0.03
0.095	0.9936	-0.0686	-0.04
0.100	0.9929	-0.0720	-0.04
0.105	0.9920	-0.0757	-0.04
0.110	0.9915	-0.0791	-0.05
0.115	0.9906	-0.0826	-0.05

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9899	-0.0860	-0.06
0.125	0.9890	-0.0896	-0.06
0.130	0.9880	-0.0929	-0.07
0.135	0.9872	-0.0964	-0.07
0.140	0.9864	-0.1000	-0.07
0.145	0.9853	-0.1034	-0.08
0.150	0.9843	-0.1067	-0.09
0.155	0.9834	-0.1101	-0.09
0.160	0.9824	-0.1134	-0.10
0.165	0.9814	-0.1168	-0.10
0.170	0.9803	-0.1201	-0.11
0.175	0.9791	-0.1235	-0.11
0.180	0.9778	-0.1266	-0.12
0.185	0.9766	-0.1300	-0.13
0.190	0.9753	-0.1334	-0.14
0.195	0.9741	-0.1364	-0.14
0.200	0.9729	-0.1398	-0.15
0.205	0.9715	-0.1431	-0.16
0.210	0.9702	-0.1462	-0.17
0.215	0.9689	-0.1494	-0.17
0.220	0.9674	-0.1524	-0.18
0.225	0.9661	-0.1556	-0.19
0.230	0.9646	-0.1588	-0.20
0.235	0.9632	-0.1620	-0.20
0.240	0.9618	-0.1650	-0.21
0.245	0.9602	-0.1680	-0.22
0.250	0.9587	-0.1711	-0.23
0.255	0.9572	-0.1740	-0.24
0.260	0.9555	-0.1771	-0.25
0.265	0.9539	-0.1800	-0.26
0.270	0.9523	-0.1831	-0.27
0.275	0.9507	-0.1860	-0.28
0.280	0.9491	-0.1888	-0.29
0.285	0.9474	-0.1918	-0.29
0.290	0.9457	-0.1946	-0.30
0.295	0.9439	-0.1974	-0.32
0.300	0.9422	-0.2002	-0.33
0.305	0.9405	-0.2029	-0.34
0.310	0.9387	-0.2057	-0.35
0.315	0.9370	-0.2085	-0.36
0.320	0.9352	-0.2112	-0.37
0.325	0.9333	-0.2139	-0.38
0.330	0.9314	-0.2165	-0.39
0.335	0.9297	-0.2191	-0.40
0.340	0.9279	-0.2217	-0.41
0.345	0.9260	-0.2243	-0.42

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9241	-0.2269	-0.43
0.355	0.9222	-0.2293	-0.44
0.360	0.9203	-0.2318	-0.45
0.365	0.9185	-0.2344	-0.46
0.370	0.9164	-0.2368	-0.48
0.375	0.9144	-0.2392	-0.49
0.380	0.9126	-0.2417	-0.50
0.385	0.9105	-0.2441	-0.51
0.390	0.9087	-0.2464	-0.52
0.395	0.9068	-0.2487	-0.53
0.400	0.9050	-0.2510	-0.55
0.405	0.9030	-0.2533	-0.56
0.410	0.9010	-0.2555	-0.57
0.415	0.8991	-0.2577	-0.58
0.420	0.8970	-0.2599	-0.59
0.425	0.8951	-0.2621	-0.61
0.430	0.8932	-0.2641	-0.62
0.435	0.8911	-0.2661	-0.63
0.440	0.8893	-0.2684	-0.64
0.445	0.8873	-0.2703	-0.65
0.450	0.8853	-0.2723	-0.67
0.455	0.8835	-0.2744	-0.68
0.460	0.8814	-0.2764	-0.69
0.465	0.8794	-0.2784	-0.70
0.470	0.8773	-0.2802	-0.72
0.475	0.8754	-0.2822	-0.73
0.480	0.8733	-0.2839	-0.74
0.485	0.8715	-0.2858	-0.75
0.490	0.8695	-0.2877	-0.76
0.495	0.8676	-0.2896	-0.77
0.500	0.8657	-0.2913	-0.79
0.505	0.8637	-0.2931	-0.80
0.510	0.8617	-0.2948	-0.81
0.515	0.8598	-0.2966	-0.82
0.520	0.8579	-0.2983	-0.84
0.525	0.8561	-0.3000	-0.85
0.530	0.8542	-0.3015	-0.86
0.535	0.8523	-0.3033	-0.87
0.540	0.8503	-0.3048	-0.88
0.545	0.8484	-0.3065	-0.90
0.550	0.8465	-0.3082	-0.91
0.555	0.8445	-0.3097	-0.92
0.560	0.8426	-0.3113	-0.93
0.565	0.8407	-0.3128	-0.94
0.570	0.8387	-0.3143	-0.96
0.575	0.8369	-0.3158	-0.97

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8350	-0.3173	-0.98
0.585	0.8331	-0.3188	-0.99
0.590	0.8312	-0.3203	-1.00
0.595	0.8293	-0.3217	-1.02
0.600	0.8274	-0.3230	-1.03
0.605	0.8255	-0.3244	-1.04
0.610	0.8236	-0.3257	-1.05
0.615	0.8218	-0.3271	-1.07
0.620	0.8199	-0.3285	-1.08
0.625	0.8179	-0.3298	-1.09
0.630	0.8161	-0.3310	-1.10
0.635	0.8143	-0.3323	-1.12
0.640	0.8124	-0.3337	-1.13
0.645	0.8105	-0.3349	-1.14
0.650	0.8088	-0.3362	-1.15
0.655	0.8069	-0.3373	-1.16
0.660	0.8051	-0.3385	-1.18
0.665	0.8032	-0.3398	-1.19
0.670	0.8013	-0.3409	-1.20
0.675	0.7994	-0.3421	-1.21
0.680	0.7978	-0.3431	-1.23
0.685	0.7959	-0.3443	-1.24
0.690	0.7940	-0.3452	-1.25
0.695	0.7922	-0.3463	-1.26
0.700	0.7903	-0.3474	-1.28
0.705	0.7883	-0.3485	-1.29
0.710	0.7867	-0.3497	-1.30
0.715	0.7848	-0.3507	-1.31
0.720	0.7829	-0.3517	-1.33
0.725	0.7810	-0.3526	-1.34
0.730	0.7791	-0.3537	-1.35
0.735	0.7775	-0.3547	-1.37
0.740	0.7754	-0.3556	-1.38
0.745	0.7736	-0.3566	-1.39
0.750	0.7718	-0.3575	-1.41
0.755	0.7701	-0.3584	-1.42
0.760	0.7681	-0.3593	-1.43
0.765	0.7663	-0.3602	-1.44
0.770	0.7644	-0.3611	-1.46
0.775	0.7625	-0.3622	-1.47
0.780	0.7606	-0.3630	-1.49
0.785	0.7587	-0.3639	-1.50
0.790	0.7570	-0.3648	-1.51
0.795	0.7550	-0.3657	-1.53
0.800	0.7533	-0.3664	-1.54
0.805	0.7512	-0.3673	-1.55

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7493	-0.3680	-1.57
0.815	0.7474	-0.3688	-1.58
0.820	0.7456	-0.3695	-1.60
0.825	0.7438	-0.3703	-1.61
0.830	0.7419	-0.3711	-1.62
0.835	0.7399	-0.3719	-1.64
0.840	0.7382	-0.3727	-1.65
0.845	0.7363	-0.3735	-1.66
0.850	0.7344	-0.3741	-1.68
0.855	0.7326	-0.3748	-1.69
0.860	0.7306	-0.3756	-1.71
0.865	0.7288	-0.3762	-1.72
0.870	0.7269	-0.3770	-1.74
0.875	0.7249	-0.3776	-1.75
0.880	0.7232	-0.3783	-1.76
0.885	0.7214	-0.3789	-1.78
0.890	0.7193	-0.3796	-1.79
0.895	0.7176	-0.3803	-1.81
0.900	0.7157	-0.3808	-1.82
0.905	0.7139	-0.3815	-1.84
0.910	0.7121	-0.3822	-1.85
0.915	0.7102	-0.3827	-1.87
0.920	0.7083	-0.3833	-1.88
0.925	0.7065	-0.3839	-1.89
0.930	0.7047	-0.3845	-1.91
0.935	0.7028	-0.3852	-1.92
0.940	0.7009	-0.3857	-1.94
0.945	0.6991	-0.3863	-1.95
0.950	0.6972	-0.3869	-1.97
0.955	0.6954	-0.3875	-1.98
0.960	0.6936	-0.3880	-2.00
0.965	0.6917	-0.3884	-2.01
0.970	0.6900	-0.3890	-2.02
0.975	0.6882	-0.3896	-2.04
0.980	0.6863	-0.3899	-2.05
0.985	0.6846	-0.3906	-2.07
0.990	0.6828	-0.3910	-2.08
0.995	0.6810	-0.3915	-2.10
1.000	0.6793	-0.3920	-2.11
1.005	0.6774	-0.3925	-2.13
1.010	0.6758	-0.3929	-2.14
1.015	0.6740	-0.3933	-2.15
1.020	0.6723	-0.3937	-2.17
1.025	0.6704	-0.3941	-2.18
1.030	0.6688	-0.3947	-2.20
1.035	0.6671	-0.3952	-2.21

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.6654	-0.3955	-2.22
1.045	0.6637	-0.3960	-2.24
1.050	0.6620	-0.3963	-2.25
1.055	0.6605	-0.3966	-2.27
1.060	0.6588	-0.3969	-2.28
1.065	0.6572	-0.3973	-2.29
1.070	0.6556	-0.3979	-2.31
1.075	0.6540	-0.3982	-2.32
1.080	0.6524	-0.3985	-2.33
1.085	0.6509	-0.3989	-2.35
1.090	0.6493	-0.3992	-2.36
1.095	0.6478	-0.3995	-2.37
1.100	0.6462	-0.3998	-2.39
1.105	0.6448	-0.4002	-2.40
1.110	0.6432	-0.4005	-2.41
1.115	0.6417	-0.4008	-2.42
1.120	0.6403	-0.4011	-2.43
1.125	0.6388	-0.4014	-2.45
1.130	0.6373	-0.4017	-2.46
1.135	0.6359	-0.4019	-2.47
1.140	0.6345	-0.4021	-2.48
1.145	0.6331	-0.4024	-2.50
1.150	0.6317	-0.4027	-2.51
1.155	0.6303	-0.4029	-2.52
1.160	0.6289	-0.4032	-2.53
1.165	0.6277	-0.4035	-2.54
1.170	0.6264	-0.4037	-2.55
1.175	0.6251	-0.4038	-2.57
1.180	0.6238	-0.4041	-2.58
1.185	0.6226	-0.4043	-2.59
1.190	0.6214	-0.4045	-2.60
1.195	0.6201	-0.4046	-2.61
1.200	0.6189	-0.4048	-2.62
1.205	0.6177	-0.4049	-2.63
1.210	0.6165	-0.4050	-2.64
1.215	0.6153	-0.4052	-2.65
1.220	0.6141	-0.4053	-2.66
1.225	0.6129	-0.4055	-2.68
1.230	0.6117	-0.4056	-2.69
1.235	0.6106	-0.4057	-2.70
1.240	0.6094	-0.4059	-2.71
1.245	0.6082	-0.4060	-2.72
1.250	0.6072	-0.4061	-2.73
1.255	0.6061	-0.4062	-2.74
1.260	0.6050	-0.4063	-2.75
1.265	0.6039	-0.4064	-2.76

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.6028	-0.4065	-2.77
1.275	0.6018	-0.4066	-2.78
1.280	0.6007	-0.4067	-2.79
1.285	0.5996	-0.4068	-2.80
1.290	0.5984	-0.4068	-2.81
1.295	0.5973	-0.4070	-2.82
1.300	0.5962	-0.4071	-2.83
1.305	0.5953	-0.4072	-2.84
1.310	0.5943	-0.4072	-2.85
1.315	0.5932	-0.4073	-2.86
1.320	0.5922	-0.4075	-2.87
1.325	0.5912	-0.4073	-2.88
1.330	0.5902	-0.4074	-2.89
1.335	0.5893	-0.4074	-2.90
1.340	0.5882	-0.4076	-2.91
1.345	0.5872	-0.4076	-2.92
1.350	0.5863	-0.4075	-2.93
1.355	0.5852	-0.4078	-2.93
1.360	0.5842	-0.4076	-2.95
1.365	0.5832	-0.4077	-2.96
1.370	0.5822	-0.4077	-2.97
1.375	0.5812	-0.4077	-2.98
1.380	0.5801	-0.4077	-2.99
1.385	0.5790	-0.4078	-3.00
1.390	0.5780	-0.4079	-3.01
1.395	0.5770	-0.4078	-3.02
1.400	0.5759	-0.4078	-3.03
1.405	0.5748	-0.4080	-3.04
1.410	0.5736	-0.4080	-3.05
1.415	0.5726	-0.4080	-3.06
1.420	0.5714	-0.4080	-3.07
1.425	0.5703	-0.4081	-3.08
1.430	0.5691	-0.4082	-3.09
1.435	0.5679	-0.4082	-3.11
1.440	0.5669	-0.4083	-3.11
1.445	0.5656	-0.4085	-3.13
1.450	0.5646	-0.4084	-3.14
1.455	0.5635	-0.4085	-3.15
1.460	0.5623	-0.4086	-3.16
1.465	0.5610	-0.4086	-3.17
1.470	0.5599	-0.4086	-3.18
1.475	0.5587	-0.4086	-3.20
1.480	0.5575	-0.4088	-3.21
1.485	0.5562	-0.4089	-3.22
1.490	0.5549	-0.4090	-3.23
1.495	0.5538	-0.4091	-3.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.5525	-0.4092	-3.25
1.505	0.5513	-0.4091	-3.27
1.510	0.5499	-0.4092	-3.28
1.515	0.5486	-0.4092	-3.29
1.520	0.5473	-0.4092	-3.31
1.525	0.5460	-0.4094	-3.32
1.530	0.5447	-0.4094	-3.33
1.535	0.5434	-0.4095	-3.34
1.540	0.5421	-0.4095	-3.36
1.545	0.5408	-0.4095	-3.37
1.550	0.5395	-0.4095	-3.38
1.555	0.5380	-0.4096	-3.40
1.560	0.5366	-0.4097	-3.41
1.565	0.5351	-0.4098	-3.43
1.570	0.5338	-0.4099	-3.44
1.575	0.5324	-0.4099	-3.45
1.580	0.5311	-0.4100	-3.47
1.585	0.5296	-0.4100	-3.48
1.590	0.5284	-0.4102	-3.49
1.595	0.5271	-0.4103	-3.51
1.600	0.5258	-0.4104	-3.52
1.605	0.5243	-0.4105	-3.53
1.610	0.5230	-0.4105	-3.55
1.615	0.5216	-0.4106	-3.56
1.620	0.5203	-0.4105	-3.57
1.625	0.5189	-0.4107	-3.59
1.630	0.5176	-0.4107	-3.60
1.635	0.5162	-0.4107	-3.61
1.640	0.5147	-0.4107	-3.63
1.645	0.5132	-0.4107	-3.64
1.650	0.5120	-0.4110	-3.66
1.655	0.5106	-0.4110	-3.67
1.660	0.5092	-0.4111	-3.68
1.665	0.5078	-0.4112	-3.70
1.670	0.5063	-0.4112	-3.71
1.675	0.5052	-0.4112	-3.72
1.680	0.5037	-0.4113	-3.74
1.685	0.5025	-0.4114	-3.75
1.690	0.5010	-0.4113	-3.77
1.695	0.4998	-0.4113	-3.78
1.700	0.4984	-0.4113	-3.79
1.705	0.4972	-0.4112	-3.81
1.710	0.4958	-0.4114	-3.82
1.715	0.4946	-0.4113	-3.83
1.720	0.4931	-0.4114	-3.85
1.725	0.4917	-0.4114	-3.86

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.4906	-0.4114	-3.87
1.735	0.4893	-0.4114	-3.89
1.740	0.4881	-0.4115	-3.90
1.745	0.4869	-0.4117	-3.91
1.750	0.4857	-0.4115	-3.92
1.755	0.4846	-0.4116	-3.93
1.760	0.4834	-0.4116	-3.95
1.765	0.4824	-0.4117	-3.96
1.770	0.4813	-0.4117	-3.97
1.775	0.4803	-0.4116	-3.98
1.780	0.4794	-0.4115	-3.99
1.785	0.4784	-0.4114	-4.00
1.790	0.4773	-0.4113	-4.01
1.795	0.4762	-0.4113	-4.02
1.800	0.4751	-0.4113	-4.04
1.805	0.4742	-0.4114	-4.04
1.810	0.4731	-0.4114	-4.06
1.815	0.4723	-0.4114	-4.06
1.820	0.4714	-0.4114	-4.07
1.825	0.4706	-0.4114	-4.08
1.830	0.4697	-0.4113	-4.09
1.835	0.4689	-0.4113	-4.10
1.840	0.4680	-0.4112	-4.11
1.845	0.4673	-0.4112	-4.12
1.850	0.4664	-0.4110	-4.13
1.855	0.4656	-0.4108	-4.14
1.860	0.4647	-0.4107	-4.15
1.865	0.4640	-0.4106	-4.16
1.870	0.4631	-0.4105	-4.17
1.875	0.4622	-0.4104	-4.18
1.880	0.4616	-0.4106	-4.18
1.885	0.4609	-0.4107	-4.19
1.890	0.4604	-0.4106	-4.20
1.895	0.4598	-0.4106	-4.20
1.900	0.4592	-0.4105	-4.21
1.905	0.4585	-0.4102	-4.22
1.910	0.4579	-0.4103	-4.23
1.915	0.4573	-0.4103	-4.23
1.920	0.4567	-0.4103	-4.24
1.925	0.4561	-0.4103	-4.24
1.930	0.4556	-0.4103	-4.25
1.935	0.4551	-0.4102	-4.26
1.940	0.4545	-0.4102	-4.26
1.945	0.4540	-0.4101	-4.27
1.950	0.4534	-0.4101	-4.27
1.955	0.4529	-0.4101	-4.28

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.4525	-0.4099	-4.29
1.965	0.4520	-0.4099	-4.29
1.970	0.4515	-0.4098	-4.30
1.975	0.4511	-0.4097	-4.30
1.980	0.4506	-0.4096	-4.31
1.985	0.4502	-0.4095	-4.31
1.990	0.4497	-0.4093	-4.32
1.995	0.4492	-0.4093	-4.33
2.000	0.4486	-0.4090	-4.34
2.005	0.4482	-0.4091	-4.34
2.010	0.4477	-0.4089	-4.35
2.015	0.4473	-0.4089	-4.35
2.020	0.4468	-0.4088	-4.36
2.025	0.4462	-0.4087	-4.36
2.030	0.4458	-0.4087	-4.37
2.035	0.4452	-0.4087	-4.37
2.040	0.4447	-0.4085	-4.38
2.045	0.4441	-0.4084	-4.39
2.050	0.4437	-0.4084	-4.39
2.055	0.4431	-0.4083	-4.40
2.060	0.4426	-0.4080	-4.41
2.065	0.4420	-0.4081	-4.41
2.070	0.4414	-0.4080	-4.42
2.075	0.4409	-0.4079	-4.43
2.080	0.4403	-0.4077	-4.44
2.085	0.4397	-0.4076	-4.44
2.090	0.4389	-0.4075	-4.45
2.095	0.4384	-0.4074	-4.46
2.100	0.4376	-0.4073	-4.47
2.105	0.4369	-0.4074	-4.48
2.110	0.4363	-0.4073	-4.48
2.115	0.4355	-0.4072	-4.49
2.120	0.4348	-0.4072	-4.50
2.125	0.4340	-0.4072	-4.51
2.130	0.4332	-0.4071	-4.52
2.135	0.4324	-0.4071	-4.53
2.140	0.4316	-0.4071	-4.53
2.145	0.4308	-0.4070	-4.54
2.150	0.4300	-0.4070	-4.55
2.155	0.4292	-0.4070	-4.56
2.160	0.4282	-0.4069	-4.57
2.165	0.4274	-0.4069	-4.58
2.170	0.4265	-0.4068	-4.59
2.175	0.4256	-0.4068	-4.60
2.180	0.4246	-0.4068	-4.61
2.185	0.4237	-0.4067	-4.62

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.4229	-0.4067	-4.63
2.195	0.4219	-0.4067	-4.64
2.200	0.4208	-0.4068	-4.65
2.205	0.4199	-0.4067	-4.66
2.210	0.4189	-0.4068	-4.67
2.215	0.4178	-0.4066	-4.69
2.220	0.4168	-0.4067	-4.70
2.225	0.4158	-0.4065	-4.71
2.230	0.4147	-0.4068	-4.72
2.235	0.4137	-0.4066	-4.73
2.240	0.4126	-0.4067	-4.74
2.245	0.4117	-0.4065	-4.75
2.250	0.4106	-0.4069	-4.76
2.255	0.4096	-0.4067	-4.77
2.260	0.4087	-0.4068	-4.78
2.265	0.4075	-0.4068	-4.79
2.270	0.4065	-0.4068	-4.81
2.275	0.4054	-0.4068	-4.82
2.280	0.4044	-0.4069	-4.83
2.285	0.4033	-0.4070	-4.84
2.290	0.4023	-0.4069	-4.85
2.295	0.4012	-0.4070	-4.86
2.300	0.4003	-0.4070	-4.87
2.305	0.3990	-0.4069	-4.88
2.310	0.3980	-0.4069	-4.89
2.315	0.3970	-0.4071	-4.90
2.320	0.3959	-0.4071	-4.92
2.325	0.3949	-0.4071	-4.93
2.330	0.3938	-0.4071	-4.94
2.335	0.3928	-0.4072	-4.95
2.340	0.3917	-0.4073	-4.96
2.345	0.3906	-0.4074	-4.97
2.350	0.3897	-0.4074	-4.98
2.355	0.3887	-0.4075	-4.99
2.360	0.3878	-0.4074	-5.00
2.365	0.3867	-0.4074	-5.01
2.370	0.3859	-0.4075	-5.02
2.375	0.3849	-0.4075	-5.03
2.380	0.3840	-0.4075	-5.04
2.385	0.3831	-0.4074	-5.05
2.390	0.3822	-0.4076	-5.05
2.395	0.3813	-0.4077	-5.06
2.400	0.3804	-0.4078	-5.07
2.405	0.3796	-0.4078	-5.08
2.410	0.3788	-0.4078	-5.09
2.415	0.3780	-0.4078	-5.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.3772	-0.4078	-5.11
2.425	0.3764	-0.4080	-5.11
2.430	0.3757	-0.4079	-5.12
2.435	0.3749	-0.4079	-5.13
2.440	0.3742	-0.4079	-5.14
2.445	0.3734	-0.4079	-5.15
2.450	0.3728	-0.4078	-5.15
2.455	0.3722	-0.4078	-5.16
2.460	0.3716	-0.4078	-5.17
2.465	0.3710	-0.4077	-5.17
2.470	0.3703	-0.4078	-5.18
2.475	0.3698	-0.4076	-5.19
2.480	0.3691	-0.4075	-5.19
2.485	0.3685	-0.4076	-5.20
2.490	0.3680	-0.4075	-5.21
2.495	0.3674	-0.4075	-5.21
2.500	0.3669	-0.4075	-5.22
2.505	0.3663	-0.4076	-5.22
2.510	0.3658	-0.4076	-5.23
2.515	0.3652	-0.4075	-5.24
2.520	0.3647	-0.4075	-5.24
2.525	0.3641	-0.4076	-5.25
2.530	0.3637	-0.4077	-5.25
2.535	0.3631	-0.4076	-5.26
2.540	0.3626	-0.4076	-5.26
2.545	0.3621	-0.4076	-5.27
2.550	0.3616	-0.4076	-5.27
2.555	0.3612	-0.4073	-5.28
2.560	0.3607	-0.4072	-5.29
2.565	0.3605	-0.4069	-5.29
2.570	0.3602	-0.4066	-5.30
2.575	0.3603	-0.4063	-5.30
2.580	0.3602	-0.4061	-5.31
2.585	0.3603	-0.4060	-5.31
2.590	0.3604	-0.4059	-5.31
2.595	0.3605	-0.4060	-5.31
2.600	0.3605	-0.4061	-5.30
2.605	0.3606	-0.4062	-5.30
2.610	0.3605	-0.4063	-5.30
2.615	0.3604	-0.4064	-5.30
2.620	0.3603	-0.4063	-5.30
2.625	0.3603	-0.4064	-5.30
2.630	0.3600	-0.4064	-5.30
2.635	0.3600	-0.4064	-5.30
2.640	0.3599	-0.4063	-5.31
2.645	0.3597	-0.4063	-5.31

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.3597	-0.4063	-5.31
2.655	0.3596	-0.4063	-5.31
2.660	0.3594	-0.4062	-5.31
2.665	0.3592	-0.4061	-5.32
2.670	0.3590	-0.4061	-5.32
2.675	0.3587	-0.4060	-5.32
2.680	0.3587	-0.4061	-5.32
2.685	0.3586	-0.4060	-5.32
2.690	0.3584	-0.4060	-5.33
2.695	0.3583	-0.4061	-5.33
2.700	0.3582	-0.4061	-5.33
2.705	0.3581	-0.4060	-5.33
2.710	0.3580	-0.4060	-5.33
2.715	0.3581	-0.4059	-5.33
2.720	0.3580	-0.4059	-5.33
2.725	0.3578	-0.4056	-5.34
2.730	0.3577	-0.4055	-5.34
2.735	0.3576	-0.4051	-5.35
2.740	0.3573	-0.4048	-5.35
2.745	0.3571	-0.4046	-5.36
2.750	0.3569	-0.4044	-5.36
2.755	0.3565	-0.4041	-5.37
2.760	0.3562	-0.4039	-5.38
2.765	0.3558	-0.4036	-5.38
2.770	0.3556	-0.4033	-5.39
2.775	0.3551	-0.4031	-5.40
2.780	0.3548	-0.4028	-5.40
2.785	0.3543	-0.4026	-5.41
2.790	0.3539	-0.4025	-5.42
2.795	0.3536	-0.4022	-5.43
2.800	0.3531	-0.4020	-5.43
2.805	0.3526	-0.4020	-5.44
2.810	0.3522	-0.4017	-5.45
2.815	0.3518	-0.4016	-5.45
2.820	0.3512	-0.4016	-5.46
2.825	0.3508	-0.4014	-5.46
2.830	0.3504	-0.4013	-5.47
2.835	0.3499	-0.4011	-5.48
2.840	0.3495	-0.4011	-5.48
2.845	0.3490	-0.4010	-5.49
2.850	0.3486	-0.4008	-5.49
2.855	0.3482	-0.4008	-5.50
2.860	0.3476	-0.4008	-5.51
2.865	0.3474	-0.4006	-5.51
2.870	0.3470	-0.4004	-5.52
2.875	0.3467	-0.4001	-5.52

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.3463	-0.3998	-5.53
2.885	0.3460	-0.3994	-5.54
2.890	0.3454	-0.3991	-5.55
2.895	0.3448	-0.3988	-5.56
2.900	0.3441	-0.3985	-5.57
2.905	0.3435	-0.3981	-5.58
2.910	0.3429	-0.3978	-5.59
2.915	0.3421	-0.3977	-5.60
2.920	0.3414	-0.3977	-5.61
2.925	0.3405	-0.3974	-5.62
2.930	0.3398	-0.3972	-5.64
2.935	0.3392	-0.3972	-5.64
2.940	0.3385	-0.3973	-5.65
2.945	0.3378	-0.3974	-5.65
2.950	0.3373	-0.3975	-5.66
2.955	0.3368	-0.3976	-5.66
2.960	0.3363	-0.3977	-5.67
2.965	0.3358	-0.3980	-5.67
2.970	0.3355	-0.3979	-5.67
2.975	0.3352	-0.3982	-5.67
2.980	0.3349	-0.3983	-5.67
2.985	0.3348	-0.3985	-5.67
2.990	0.3347	-0.3984	-5.67
2.995	0.3347	-0.3985	-5.67
3.000	0.3346	-0.3984	-5.68
3.005	0.3327	-0.3981	-5.70
3.010	0.3326	-0.3983	-5.70
3.015	0.3326	-0.3985	-5.70
3.020	0.3326	-0.3986	-5.70
3.025	0.3326	-0.3987	-5.69
3.030	0.3328	-0.3987	-5.69
3.035	0.3329	-0.3986	-5.69
3.040	0.3331	-0.3983	-5.69
3.045	0.3331	-0.3980	-5.70
3.050	0.3333	-0.3975	-5.70
3.055	0.3334	-0.3972	-5.70
3.060	0.3333	-0.3967	-5.71
3.065	0.3330	-0.3960	-5.72
3.070	0.3328	-0.3956	-5.73
3.075	0.3324	-0.3950	-5.74
3.080	0.3321	-0.3945	-5.75
3.085	0.3316	-0.3940	-5.76
3.090	0.3310	-0.3935	-5.78
3.095	0.3305	-0.3933	-5.79
3.100	0.3299	-0.3932	-5.79
3.105	0.3294	-0.3931	-5.80

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.3287	-0.3931	-5.81
3.115	0.3281	-0.3929	-5.82
3.120	0.3275	-0.3931	-5.82
3.125	0.3270	-0.3934	-5.82
3.130	0.3267	-0.3937	-5.82
3.135	0.3263	-0.3940	-5.82
3.140	0.3260	-0.3944	-5.82
3.145	0.3258	-0.3950	-5.82
3.150	0.3257	-0.3953	-5.81
3.155	0.3258	-0.3955	-5.81
3.160	0.3258	-0.3960	-5.80
3.165	0.3260	-0.3963	-5.80
3.170	0.3262	-0.3965	-5.79
3.175	0.3264	-0.3969	-5.78
3.180	0.3267	-0.3970	-5.78
3.185	0.3271	-0.3969	-5.78
3.190	0.3273	-0.3968	-5.77
3.195	0.3277	-0.3964	-5.78
3.200	0.3280	-0.3961	-5.78
3.205	0.3280	-0.3957	-5.78
3.210	0.3282	-0.3951	-5.79
3.215	0.3281	-0.3944	-5.80
3.220	0.3279	-0.3940	-5.81
3.225	0.3275	-0.3935	-5.82
3.230	0.3269	-0.3934	-5.82
3.235	0.3264	-0.3935	-5.83
3.240	0.3261	-0.3936	-5.83
3.245	0.3260	-0.3937	-5.83
3.250	0.3256	-0.3937	-5.83
3.255	0.3255	-0.3939	-5.83
3.260	0.3253	-0.3942	-5.83
3.265	0.3251	-0.3944	-5.83
3.270	0.3250	-0.3945	-5.83
3.275	0.3250	-0.3950	-5.82
3.280	0.3248	-0.3949	-5.83
3.285	0.3248	-0.3952	-5.82
3.290	0.3247	-0.3955	-5.82
3.295	0.3247	-0.3957	-5.82
3.300	0.3246	-0.3957	-5.82
3.305	0.3246	-0.3958	-5.82
3.310	0.3245	-0.3958	-5.82
3.315	0.3246	-0.3959	-5.82
3.320	0.3247	-0.3960	-5.81
3.325	0.3246	-0.3963	-5.81
3.330	0.3247	-0.3961	-5.81
3.335	0.3247	-0.3959	-5.81

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.3248	-0.3960	-5.81
3.345	0.3248	-0.3958	-5.81
3.350	0.3249	-0.3956	-5.82
3.355	0.3249	-0.3954	-5.82
3.360	0.3249	-0.3951	-5.82
3.365	0.3249	-0.3949	-5.83
3.370	0.3248	-0.3945	-5.83
3.375	0.3246	-0.3942	-5.84
3.380	0.3245	-0.3938	-5.84
3.385	0.3242	-0.3935	-5.85
3.390	0.3241	-0.3929	-5.86
3.395	0.3237	-0.3926	-5.87
3.400	0.3234	-0.3923	-5.88
3.405	0.3232	-0.3915	-5.89
3.410	0.3228	-0.3910	-5.90
3.415	0.3222	-0.3906	-5.91
3.420	0.3219	-0.3904	-5.92
3.425	0.3214	-0.3899	-5.93
3.430	0.3206	-0.3896	-5.94
3.435	0.3200	-0.3890	-5.96
3.440	0.3193	-0.3888	-5.97
3.445	0.3185	-0.3884	-5.98
3.450	0.3177	-0.3882	-5.99
3.455	0.3170	-0.3880	-6.00
3.460	0.3163	-0.3880	-6.01
3.465	0.3156	-0.3878	-6.02
3.470	0.3148	-0.3879	-6.03
3.475	0.3142	-0.3880	-6.03
3.480	0.3134	-0.3881	-6.04
3.485	0.3128	-0.3881	-6.05
3.490	0.3124	-0.3888	-6.04
3.495	0.3119	-0.3888	-6.05
3.500	0.3118	-0.3891	-6.05
3.505	0.3115	-0.3893	-6.04
3.510	0.3116	-0.3894	-6.04
3.515	0.3117	-0.3894	-6.04
3.520	0.3118	-0.3894	-6.04
3.525	0.3120	-0.3893	-6.04
3.530	0.3122	-0.3889	-6.04
3.535	0.3130	-0.3882	-6.04
3.540	0.3132	-0.3873	-6.05
3.545	0.3134	-0.3861	-6.07
3.550	0.3131	-0.3847	-6.09
3.555	0.3129	-0.3832	-6.11
3.560	0.3121	-0.3817	-6.14
3.565	0.3111	-0.3801	-6.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.3097	-0.3786	-6.21
3.575	0.3076	-0.3765	-6.26
3.580	0.3056	-0.3760	-6.29
3.585	0.3032	-0.3754	-6.33
3.590	0.3009	-0.3762	-6.34
3.595	0.2987	-0.3770	-6.36
3.600	0.2972	-0.3784	-6.35
3.605	0.2961	-0.3799	-6.35
3.610	0.2958	-0.3815	-6.33
3.615	0.2961	-0.3828	-6.30
3.620	0.2967	-0.3838	-6.28
3.625	0.2975	-0.3845	-6.26
3.630	0.2985	-0.3845	-6.25
3.635	0.2996	-0.3841	-6.25
3.640	0.2999	-0.3834	-6.25
3.645	0.3004	-0.3828	-6.26
3.650	0.3006	-0.3819	-6.27
3.655	0.3006	-0.3811	-6.28
3.660	0.3005	-0.3802	-6.29
3.665	0.3002	-0.3795	-6.31
3.670	0.3000	-0.3786	-6.32
3.675	0.2996	-0.3779	-6.33
3.680	0.2995	-0.3770	-6.35
3.685	0.2989	-0.3763	-6.36
3.690	0.2983	-0.3755	-6.38
3.695	0.2976	-0.3747	-6.40
3.700	0.2968	-0.3738	-6.42
3.705	0.2956	-0.3731	-6.45
3.710	0.2944	-0.3726	-6.47
3.715	0.2930	-0.3721	-6.49
3.720	0.2914	-0.3719	-6.51
3.725	0.2898	-0.3722	-6.53
3.730	0.2876	-0.3724	-6.55
3.735	0.2862	-0.3734	-6.55
3.740	0.2848	-0.3749	-6.54
3.745	0.2841	-0.3763	-6.53
3.750	0.2837	-0.3783	-6.51
3.755	0.2838	-0.3798	-6.48
3.760	0.2847	-0.3815	-6.45
3.765	0.2861	-0.3828	-6.41
3.770	0.2881	-0.3832	-6.39
3.775	0.2901	-0.3838	-6.36
3.780	0.2919	-0.3809	-6.38
3.785	0.2930	-0.3794	-6.39
3.790	0.2927	-0.3773	-6.42
3.795	0.2919	-0.3755	-6.45

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.2908	-0.3742	-6.49
3.805	0.2893	-0.3736	-6.51
3.810	0.2881	-0.3731	-6.53
3.815	0.2868	-0.3726	-6.55
3.820	0.2855	-0.3727	-6.57
3.825	0.2843	-0.3730	-6.58
3.830	0.2831	-0.3734	-6.58
3.835	0.2826	-0.3759	-6.55
3.840	0.2822	-0.3762	-6.55
3.845	0.2821	-0.3763	-6.55
3.850	0.2816	-0.3764	-6.56
3.855	0.2814	-0.3766	-6.56
3.860	0.2811	-0.3766	-6.56
3.865	0.2808	-0.3765	-6.56
3.870	0.2803	-0.3766	-6.57
3.875	0.2800	-0.3766	-6.57
3.880	0.2796	-0.3766	-6.58
3.885	0.2791	-0.3767	-6.58
3.890	0.2787	-0.3768	-6.58
3.895	0.2783	-0.3767	-6.59
3.900	0.2777	-0.3769	-6.59
3.905	0.2772	-0.3768	-6.60
3.910	0.2767	-0.3769	-6.60
3.915	0.2761	-0.3769	-6.61
3.920	0.2754	-0.3772	-6.61
3.925	0.2749	-0.3771	-6.62
3.930	0.2743	-0.3770	-6.63
3.935	0.2738	-0.3772	-6.63
3.940	0.2731	-0.3772	-6.64
3.945	0.2725	-0.3771	-6.65
3.950	0.2717	-0.3772	-6.65
3.955	0.2712	-0.3772	-6.66
3.960	0.2704	-0.3773	-6.67
3.965	0.2698	-0.3773	-6.67
3.970	0.2690	-0.3772	-6.68
3.975	0.2684	-0.3772	-6.69
3.980	0.2676	-0.3772	-6.70
3.985	0.2667	-0.3772	-6.71
3.990	0.2660	-0.3772	-6.72
3.995	0.2651	-0.3771	-6.73
4.000	0.2643	-0.3772	-6.73
4.005	0.2635	-0.3771	-6.74
4.010	0.2628	-0.3772	-6.75
4.015	0.2619	-0.3769	-6.76
4.020	0.2610	-0.3769	-6.77
4.025	0.2603	-0.3771	-6.78

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.2595	-0.3769	-6.79
4.035	0.2585	-0.3769	-6.80
4.040	0.2576	-0.3769	-6.81
4.045	0.2567	-0.3767	-6.82
4.050	0.2561	-0.3767	-6.83
4.055	0.2549	-0.3764	-6.85
4.060	0.2542	-0.3765	-6.85
4.065	0.2534	-0.3765	-6.86
4.070	0.2524	-0.3762	-6.88
4.075	0.2515	-0.3762	-6.89
4.080	0.2507	-0.3762	-6.90
4.085	0.2497	-0.3760	-6.91
4.090	0.2489	-0.3759	-6.92
4.095	0.2482	-0.3759	-6.93
4.100	0.2472	-0.3759	-6.94
4.105	0.2461	-0.3759	-6.95
4.110	0.2454	-0.3761	-6.95
4.115	0.2445	-0.3760	-6.96
4.120	0.2436	-0.3762	-6.97
4.125	0.2426	-0.3763	-6.98
4.130	0.2415	-0.3764	-6.99
4.135	0.2404	-0.3765	-7.00
4.140	0.2395	-0.3765	-7.01
4.145	0.2381	-0.3765	-7.02
4.150	0.2370	-0.3764	-7.04
4.155	0.2361	-0.3762	-7.05
4.160	0.2351	-0.3760	-7.06
4.165	0.2344	-0.3760	-7.07
4.170	0.2334	-0.3758	-7.08
4.175	0.2327	-0.3758	-7.09
4.180	0.2320	-0.3755	-7.10
4.185	0.2314	-0.3753	-7.11
4.190	0.2309	-0.3751	-7.12
4.195	0.2302	-0.3751	-7.13
4.200	0.2297	-0.3750	-7.13
4.205	0.2293	-0.3752	-7.14
4.210	0.2287	-0.3751	-7.14
4.215	0.2283	-0.3752	-7.15
4.220	0.2278	-0.3751	-7.15
4.225	0.2274	-0.3753	-7.16
4.230	0.2271	-0.3753	-7.16
4.235	0.2268	-0.3753	-7.16
4.240	0.2264	-0.3754	-7.16
4.245	0.2261	-0.3756	-7.16
4.250	0.2259	-0.3756	-7.16
4.255	0.2257	-0.3758	-7.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	0.2253	-0.3759	-7.17
4.265	0.2252	-0.3761	-7.16
4.270	0.2250	-0.3763	-7.16
4.275	0.2249	-0.3764	-7.16
4.280	0.2246	-0.3766	-7.16
4.285	0.2245	-0.3768	-7.16
4.290	0.2244	-0.3767	-7.16
4.295	0.2242	-0.3769	-7.16
4.300	0.2241	-0.3770	-7.16
4.305	0.2239	-0.3772	-7.16
4.310	0.2238	-0.3772	-7.16
4.315	0.2237	-0.3773	-7.16
4.320	0.2235	-0.3774	-7.16
4.325	0.2234	-0.3775	-7.16
4.330	0.2232	-0.3778	-7.15
4.335	0.2230	-0.3780	-7.15
4.340	0.2228	-0.3784	-7.15
4.345	0.2229	-0.3786	-7.14
4.350	0.2228	-0.3789	-7.14
4.355	0.2227	-0.3793	-7.13
4.360	0.2227	-0.3797	-7.13
4.365	0.2227	-0.3801	-7.12
4.370	0.2229	-0.3804	-7.11
4.375	0.2231	-0.3809	-7.10
4.380	0.2234	-0.3812	-7.10
4.385	0.2236	-0.3815	-7.09
4.390	0.2239	-0.3818	-7.08
4.395	0.2242	-0.3821	-7.07
4.400	0.2245	-0.3824	-7.06
4.405	0.2249	-0.3826	-7.06
4.410	0.2252	-0.3827	-7.05
4.415	0.2255	-0.3826	-7.05
4.420	0.2258	-0.3827	-7.05
4.425	0.2264	-0.3826	-7.04
4.430	0.2265	-0.3827	-7.04
4.435	0.2267	-0.3826	-7.04
4.440	0.2267	-0.3826	-7.04
4.445	0.2271	-0.3825	-7.04
4.450	0.2271	-0.3827	-7.03
4.455	0.2274	-0.3826	-7.03
4.460	0.2276	-0.3827	-7.03
4.465	0.2279	-0.3830	-7.02
4.470	0.2283	-0.3829	-7.02
4.475	0.2288	-0.3834	-7.00
4.480	0.2295	-0.3837	-6.99
4.485	0.2299	-0.3843	-6.98

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	0.2304	-0.3849	-6.96
4.495	0.2308	-0.3856	-6.95
4.500	0.2314	-0.3861	-6.93
4.505	0.2319	-0.3865	-6.92
4.510	0.2325	-0.3868	-6.91
4.515	0.2330	-0.3871	-6.90
4.520	0.2338	-0.3872	-6.89
4.525	0.2342	-0.3871	-6.89
4.530	0.2347	-0.3871	-6.88
4.535	0.2351	-0.3868	-6.88
4.540	0.2352	-0.3868	-6.88
4.545	0.2355	-0.3866	-6.88
4.550	0.2356	-0.3866	-6.88
4.555	0.2358	-0.3865	-6.88
4.560	0.2360	-0.3864	-6.88
4.565	0.2361	-0.3862	-6.88
4.570	0.2361	-0.3862	-6.88
4.575	0.2363	-0.3860	-6.89
4.580	0.2363	-0.3858	-6.89
4.585	0.2364	-0.3857	-6.89
4.590	0.2364	-0.3854	-6.90
4.595	0.2364	-0.3852	-6.90
4.600	0.2363	-0.3850	-6.90
4.605	0.2361	-0.3847	-6.91
4.610	0.2361	-0.3846	-6.91
4.615	0.2358	-0.3844	-6.92
4.620	0.2355	-0.3843	-6.92
4.625	0.2355	-0.3842	-6.92
4.630	0.2355	-0.3841	-6.93
4.635	0.2353	-0.3838	-6.93
4.640	0.2351	-0.3836	-6.94
4.645	0.2350	-0.3833	-6.94
4.650	0.2349	-0.3831	-6.95
4.655	0.2347	-0.3833	-6.95
4.660	0.2347	-0.3829	-6.95
4.665	0.2349	-0.3830	-6.95
4.670	0.2347	-0.3826	-6.96
4.675	0.2349	-0.3823	-6.96
4.680	0.2348	-0.3821	-6.96
4.685	0.2349	-0.3821	-6.96
4.690	0.2348	-0.3819	-6.97
4.695	0.2350	-0.3816	-6.97
4.700	0.2350	-0.3815	-6.97
4.705	0.2350	-0.3813	-6.98
4.710	0.2351	-0.3805	-6.99
4.715	0.2351	-0.3805	-6.99

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.2349	-0.3800	-7.00
4.725	0.2346	-0.3797	-7.01
4.730	0.2345	-0.3792	-7.02
4.735	0.2339	-0.3787	-7.03
4.740	0.2336	-0.3784	-7.04
4.745	0.2331	-0.3782	-7.05
4.750	0.2325	-0.3779	-7.06
4.755	0.2318	-0.3775	-7.07
4.760	0.2310	-0.3773	-7.08
4.765	0.2303	-0.3772	-7.09
4.770	0.2298	-0.3772	-7.10
4.775	0.2295	-0.3770	-7.10
4.780	0.2290	-0.3768	-7.11
4.785	0.2288	-0.3769	-7.11
4.790	0.2288	-0.3766	-7.12
4.795	0.2286	-0.3764	-7.12
4.800	0.2286	-0.3761	-7.13
4.805	0.2289	-0.3759	-7.13
4.810	0.2289	-0.3753	-7.14
4.815	0.2289	-0.3750	-7.14
4.820	0.2289	-0.3743	-7.16
4.825	0.2290	-0.3739	-7.16
4.830	0.2289	-0.3734	-7.17
4.835	0.2290	-0.3729	-7.18
4.840	0.2291	-0.3725	-7.18
4.845	0.2291	-0.3722	-7.19
4.850	0.2290	-0.3720	-7.19
4.855	0.2291	-0.3719	-7.19
4.860	0.2289	-0.3717	-7.20
4.865	0.2287	-0.3717	-7.20
4.870	0.2287	-0.3716	-7.20
4.875	0.2286	-0.3716	-7.20
4.880	0.2285	-0.3716	-7.20
4.885	0.2284	-0.3717	-7.20
4.890	0.2281	-0.3720	-7.20
4.895	0.2284	-0.3722	-7.20
4.900	0.2284	-0.3723	-7.20
4.905	0.2282	-0.3724	-7.19
4.910	0.2282	-0.3727	-7.19
4.915	0.2282	-0.3729	-7.19
4.920	0.2282	-0.3732	-7.18
4.925	0.2282	-0.3735	-7.18
4.930	0.2281	-0.3737	-7.17
4.935	0.2281	-0.3740	-7.17
4.940	0.2280	-0.3743	-7.17
4.945	0.2279	-0.3746	-7.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.2278	-0.3750	-7.16
4.955	0.2277	-0.3753	-7.15
4.960	0.2277	-0.3753	-7.15
4.965	0.2275	-0.3757	-7.15
4.970	0.2274	-0.3760	-7.14
4.975	0.2273	-0.3763	-7.14
4.980	0.2271	-0.3765	-7.14
4.985	0.2269	-0.3768	-7.13
4.990	0.2269	-0.3770	-7.13
4.995	0.2268	-0.3772	-7.13
5.000	0.2267	-0.3774	-7.13
5.005	0.2266	-0.3777	-7.12
5.010	0.2264	-0.3777	-7.12
5.015	0.2263	-0.3779	-7.12
5.020	0.2264	-0.3780	-7.12
5.025	0.2263	-0.3780	-7.12
5.030	0.2264	-0.3778	-7.12
5.035	0.2267	-0.3777	-7.12
5.040	0.2270	-0.3776	-7.12
5.045	0.2274	-0.3775	-7.12
5.050	0.2278	-0.3774	-7.11
5.055	0.2282	-0.3773	-7.11
5.060	0.2286	-0.3772	-7.11
5.065	0.2290	-0.3774	-7.10
5.070	0.2296	-0.3770	-7.10
5.075	0.2299	-0.3772	-7.10
5.080	0.2303	-0.3769	-7.10
5.085	0.2305	-0.3769	-7.10
5.090	0.2310	-0.3767	-7.09
5.095	0.2311	-0.3766	-7.10
5.100	0.2312	-0.3767	-7.09
5.105	0.2313	-0.3765	-7.09
5.110	0.2314	-0.3767	-7.09
5.115	0.2314	-0.3765	-7.09
5.120	0.2315	-0.3766	-7.09
5.125	0.2315	-0.3766	-7.09
5.130	0.2316	-0.3767	-7.09
5.135	0.2316	-0.3765	-7.09
5.140	0.2316	-0.3765	-7.09
5.145	0.2316	-0.3765	-7.09
5.150	0.2315	-0.3763	-7.10
5.155	0.2314	-0.3762	-7.10
5.160	0.2313	-0.3762	-7.10
5.165	0.2312	-0.3760	-7.10
5.170	0.2310	-0.3759	-7.11
5.175	0.2308	-0.3757	-7.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.2306	-0.3755	-7.12
5.185	0.2303	-0.3753	-7.12
5.190	0.2300	-0.3750	-7.13
5.195	0.2297	-0.3747	-7.14
5.200	0.2293	-0.3745	-7.15
5.205	0.2290	-0.3741	-7.16
5.210	0.2285	-0.3739	-7.17
5.215	0.2280	-0.3735	-7.18
5.220	0.2278	-0.3732	-7.19
5.225	0.2273	-0.3728	-7.20
5.230	0.2267	-0.3724	-7.21
5.235	0.2263	-0.3720	-7.22
5.240	0.2257	-0.3716	-7.23
5.245	0.2253	-0.3710	-7.25
5.250	0.2248	-0.3706	-7.26
5.255	0.2242	-0.3701	-7.28
5.260	0.2234	-0.3697	-7.29
5.265	0.2227	-0.3691	-7.31
5.270	0.2222	-0.3686	-7.32
5.275	0.2217	-0.3683	-7.33
5.280	0.2208	-0.3679	-7.35
5.285	0.2201	-0.3675	-7.36
5.290	0.2196	-0.3671	-7.38
5.295	0.2187	-0.3667	-7.39
5.300	0.2182	-0.3664	-7.40
5.305	0.2174	-0.3663	-7.41
5.310	0.2169	-0.3655	-7.43
5.315	0.2163	-0.3655	-7.44
5.320	0.2157	-0.3652	-7.45
5.325	0.2150	-0.3648	-7.47
5.330	0.2145	-0.3645	-7.47
5.335	0.2141	-0.3640	-7.49
5.340	0.2134	-0.3638	-7.50
5.345	0.2129	-0.3635	-7.51
5.350	0.2123	-0.3629	-7.53
5.355	0.2118	-0.3624	-7.54
5.360	0.2113	-0.3619	-7.55
5.365	0.2105	-0.3614	-7.57
5.370	0.2100	-0.3607	-7.59
5.375	0.2093	-0.3604	-7.60
5.380	0.2084	-0.3597	-7.62
5.385	0.2074	-0.3594	-7.64
5.390	0.2068	-0.3590	-7.65
5.395	0.2058	-0.3586	-7.67
5.400	0.2051	-0.3582	-7.69
5.405	0.2041	-0.3581	-7.70

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.2034	-0.3579	-7.71
5.415	0.2028	-0.3577	-7.72
5.420	0.2018	-0.3573	-7.74
5.425	0.2007	-0.3572	-7.75
5.430	0.1999	-0.3569	-7.76
5.435	0.1993	-0.3569	-7.77
5.440	0.1986	-0.3567	-7.78
5.445	0.1978	-0.3566	-7.79
5.450	0.1972	-0.3564	-7.80
5.455	0.1966	-0.3562	-7.81
5.460	0.1961	-0.3561	-7.82
5.465	0.1954	-0.3558	-7.83
5.470	0.1949	-0.3556	-7.84
5.475	0.1943	-0.3553	-7.85
5.480	0.1937	-0.3550	-7.86
5.485	0.1929	-0.3547	-7.88
5.490	0.1922	-0.3545	-7.89
5.495	0.1915	-0.3543	-7.90
5.500	0.1907	-0.3540	-7.91
5.505	0.1897	-0.3538	-7.93
5.510	0.1890	-0.3540	-7.93
5.515	0.1881	-0.3540	-7.94
5.520	0.1874	-0.3540	-7.95
5.525	0.1869	-0.3541	-7.95
5.530	0.1862	-0.3544	-7.95
5.535	0.1857	-0.3544	-7.96
5.540	0.1851	-0.3547	-7.96
5.545	0.1846	-0.3547	-7.96
5.550	0.1842	-0.3549	-7.96
5.555	0.1837	-0.3550	-7.97
5.560	0.1832	-0.3551	-7.97
5.565	0.1827	-0.3552	-7.97
5.570	0.1821	-0.3553	-7.98
5.575	0.1816	-0.3555	-7.98
5.580	0.1811	-0.3556	-7.98
5.585	0.1804	-0.3557	-7.98
5.590	0.1798	-0.3558	-7.99
5.595	0.1792	-0.3561	-7.99
5.600	0.1785	-0.3564	-7.99
5.605	0.1778	-0.3568	-7.99
5.610	0.1773	-0.3571	-7.99
5.615	0.1766	-0.3576	-7.98
5.620	0.1761	-0.3581	-7.98
5.625	0.1755	-0.3586	-7.98
5.630	0.1752	-0.3592	-7.97
5.635	0.1749	-0.3598	-7.96

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.1747	-0.3602	-7.95
5.645	0.1744	-0.3607	-7.94
5.650	0.1740	-0.3613	-7.94
5.655	0.1738	-0.3616	-7.93
5.660	0.1734	-0.3620	-7.93
5.665	0.1732	-0.3623	-7.92
5.670	0.1730	-0.3630	-7.91
5.675	0.1727	-0.3634	-7.91
5.680	0.1725	-0.3640	-7.90
5.685	0.1721	-0.3642	-7.90
5.690	0.1719	-0.3645	-7.89
5.695	0.1715	-0.3649	-7.89
5.700	0.1715	-0.3654	-7.88
5.705	0.1715	-0.3658	-7.87
5.710	0.1713	-0.3662	-7.87
5.715	0.1712	-0.3666	-7.86
5.720	0.1710	-0.3668	-7.86
5.725	0.1710	-0.3673	-7.85
5.730	0.1709	-0.3675	-7.84
5.735	0.1708	-0.3680	-7.84
5.740	0.1706	-0.3683	-7.83
5.745	0.1707	-0.3687	-7.82
5.750	0.1706	-0.3688	-7.82
5.755	0.1706	-0.3688	-7.82
5.760	0.1705	-0.3691	-7.82
5.765	0.1704	-0.3693	-7.81
5.770	0.1704	-0.3693	-7.81
5.775	0.1704	-0.3693	-7.81
5.780	0.1702	-0.3694	-7.81
5.785	0.1701	-0.3694	-7.82
5.790	0.1700	-0.3693	-7.82
5.795	0.1699	-0.3693	-7.82
5.800	0.1698	-0.3691	-7.82
5.805	0.1697	-0.3691	-7.82
5.810	0.1696	-0.3690	-7.83
5.815	0.1694	-0.3687	-7.83
5.820	0.1692	-0.3685	-7.84
5.825	0.1691	-0.3683	-7.85
5.830	0.1690	-0.3681	-7.85
5.835	0.1686	-0.3677	-7.86
5.840	0.1683	-0.3676	-7.87
5.845	0.1681	-0.3672	-7.88
5.850	0.1677	-0.3668	-7.89
5.855	0.1675	-0.3668	-7.89
5.860	0.1672	-0.3664	-7.90
5.865	0.1668	-0.3661	-7.91

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.1663	-0.3657	-7.92
5.875	0.1658	-0.3652	-7.94
5.880	0.1654	-0.3650	-7.94
5.885	0.1648	-0.3645	-7.96
5.890	0.1644	-0.3645	-7.96
5.895	0.1640	-0.3639	-7.98
5.900	0.1634	-0.3637	-7.99
5.905	0.1630	-0.3636	-7.99
5.910	0.1625	-0.3635	-8.00
5.915	0.1622	-0.3633	-8.00
5.920	0.1618	-0.3630	-8.02
5.925	0.1614	-0.3628	-8.02
5.930	0.1612	-0.3624	-8.03
5.935	0.1609	-0.3618	-8.05
5.940	0.1603	-0.3615	-8.06
5.945	0.1597	-0.3612	-8.07
5.950	0.1595	-0.3605	-8.09
5.955	0.1588	-0.3599	-8.10
5.960	0.1583	-0.3593	-8.12
5.965	0.1575	-0.3587	-8.14
5.970	0.1566	-0.3585	-8.15
5.975	0.1557	-0.3581	-8.17
5.980	0.1549	-0.3582	-8.17
5.985	0.1545	-0.3581	-8.18
5.990	0.1536	-0.3579	-8.19
5.995	0.1529	-0.3580	-8.20
6.000	0.1522	-0.3580	-8.20

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.